

*The lights are not magic.
Somebody is doing the work.*

How the Lights Stay On



*Why electricity has to be made and used in the same
instant — and everything that follows*

MEIRA DAVIDSON

How the Lights Stay On

Why electricity has to be made and used in the same instant — and
everything that follows

By Meira Davidson

Copyright © 2026 Meira Davidson. All rights reserved.

Set in Iowan Old Style and Helvetica Neue.

For the Jessica family kids — be curious, stay open, be kind.

Contents



	<i>Author's Note</i>	5
1.	<i>The Blink</i>	8
2.	<i>The Juggling Act</i>	14
3.	<i>The Long Road</i>	21
4.	<i>Seven Minutes in Ohio</i>	29
5.	<i>The Last Hundred Feet</i>	36
6.	<i>The Sun Doesn't Ask Permission</i>	44
7.	<i>Catching Lightning in a Bottle</i>	53
8.	<i>The Frozen Texan</i>	63
9.	<i>The Bare Roof</i>	72
10.	<i>The Atom Comes Back</i>	81
11.	<i>The Grid That Has to Grow</i>	93
12.	<i>The Grid Grows a Brain</i>	104
13.	<i>The Order of Operations</i>	116
14.	<i>The Trilemma</i>	126
15.	<i>The Lights in Shanghai</i>	136
16.	<i>The People Who Watch the Grid</i>	143
17.	<i>The Grid You'll Build</i>	154

Author's Note



This book is for anyone — kid, teenager, or curious adult — who has ever wondered how electricity actually works, and has been frustrated by the fact that most explanations stop right where it starts to get interesting.

It is also a book for people who do not yet know they want to understand the electricity grid, but who suspect they should. The grid is one of the most consequential machines humans have ever built. The lights coming on when you flip a switch is the result of an extraordinary amount of work — engineering, planning, regulation, labor, judgment — most of which is invisible. We do not, generally, teach kids how the grid works, even though they will spend their entire lives depending on it. This book is my attempt to fix that for the people who pick it up.

The book is honest about trade-offs. There is a kind of energy writing for young readers that frames the world as good guys versus bad guys — clean energy good, fossil fuels bad, certain politicians wise and others reckless. That is not the kind of writing I have tried to do. Every choice in the grid is a trade-off between things people genuinely want. Cheaper electricity often means dirtier electricity, or less reliable electricity, or both. Cleaner electricity often means more

expensive electricity, or trickier intermittency problems, or both. There is no electricity system in the world that has solved all of these problems at once. I have tried to tell you about the trade-offs honestly, without picking a side. You are old enough — and smart enough — to pick your own side.

The book is also honest about what it does not know. The energy transition is unfolding while I write this. Some facts in this book — the status of a particular policy, the deployment count of a particular technology, the latest decision from a state regulator — will be out of date by the time you read it. I have tried to be careful with these facts and to flag the places where things are changing fast. The framework should hold up. The framework, in fact, is the most important thing this book is trying to give you.

If you are reading this in middle school or high school, some of the choices being made right now about the grid will not be felt fully until you are an adult. You will inherit the grid that today's adults are building. You will also, eventually, build the grid for the next generation. The book ends with a list of unsolved problems that I genuinely think some of you will spend your careers on. If that turns out to be true, I will consider the book to have done its job.

A few practical notes.

The book is structured to be read in order, because each chapter builds on the ones before it. But chapters can also be read individually if you are curious about a specific topic. Each chapter is roughly self-contained, with references back to earlier chapters where they matter.

There are no exercises, no quizzes, no questions at the end of each chapter. This is not a textbook. It is a book.

If you find errors — and you will — I would love to know about them. The grid is changing too fast for any one book to keep up.

I hope you enjoy it.

Now go meet the Blink.

CHAPTER ONE

The Blink



On the afternoon of August 14, 2003, the lights went out across the northeastern United States and parts of Canada. They went out fast — within seven minutes, fifty million people had lost power.

In New York City, the elevators stopped between floors. The computers blanked. The air conditioning died and offices got hot quickly. People wandered out into the streets and started walking. Some walked across the Brooklyn Bridge in the August heat, dress shoes in hand, bottled water passed from stranger to stranger.

And under the streets, in the subway tunnels, the trains had stopped between stations.

That was the scene that played on every television screen for the next two days. Subway cars halted in the dark, hundreds of passengers inside, the air growing close. New York City police officers arrived at the emergency shafts — the ventilation wells that come up through the sidewalks — and lowered themselves down into the tunnels with flashlights. Then, car by car, train by train, they led people out. Office workers in suits and dress shirts. Women in skirts and heels. Tourists with shopping bags. They walked single file along the dark tracks, careful of the third rail, with a police officer's

flashlight bobbing somewhere ahead. They climbed up ladders inside the ventilation wells, hand over hand, and emerged through grates in the sidewalks of Manhattan, blinking in the late-afternoon sun, while crowds gathered on the curbs to help them up. Heels in one hand, briefcase in the other.

You did not have to be in New York that day to remember it. The whole country watched.

At an office where a friend of mine worked, the AC gave up and the staff started to leave — but how were they going to get home? The trains were not running. The buses were stuck in traffic that no traffic light could untangle. There was one car in the parking lot. It belonged to the office boy, a young guy who drove in every day because he ran the company's errands. So he drove everyone home, one by one — across boroughs, into New Jersey, out to the suburbs — for hours and hours, dropping off the executives and the accountants and the receptionists at their doorsteps as the sun went down on a city that could not turn on its lights.

By the time he finished, it was the middle of the night. He had become, for one strange afternoon, the entire transportation system of his small piece of the world.

That blackout is a story we'll come back to. We'll trace it minute by minute later in this book — what caused it, why it spread so fast, how a tree branch in Ohio could darken New York City. But right now I want to start somewhere much smaller.

I want to start inside an apartment.

A few years later I lived on the New Jersey side of the Hudson River, in an old apartment building in Hoboken. The building was old enough that the wiring inside the walls had not really been designed for what people now plug into them.

Here is what happened every August night.

I would come home, sweaty, and turn on the air conditioner. There would be a hum from the window unit as the compressor kicked on. And in the exact moment that hum started, the bulbs in my apartment would dim — just for an instant — and then come back to full brightness.

A blink. So fast you'd miss it if you weren't looking. Almost too fast to be sure you saw.

But once I noticed, I could not unsee it. Every night, same thing. The AC clicks on, the bulbs dip, the bulbs recover.

For a long time, I figured this was just a quirk of an old building. Something to roll my eyes at and forget. Then I started reading about how the grid works, and I realized something surprising:

The Blink in my Hoboken apartment, and the Blackout that took down fifty million people, were the same thing. The same physics. The same kind of failure-in-miniature. Just at different scales.

That is what this book is about.

To see why, we need to spend a minute on what is actually happening when your bulbs dim.

An air conditioner is mostly a motor. Inside the compressor, there is a heavy metal shaft that has to start spinning every time the AC clicks on. Getting a stopped motor spinning takes a *lot* of electricity for a very brief moment — usually somewhere between five and seven times the normal running current. This sudden gulp of electricity has a name. Engineers call it **inrush current**.

The wires in your wall are not magic. They look like simple metal, but they have a tiny amount of resistance, the way a garden hose has a tiny amount of friction. When a sudden spike of current rushes through a resistive wire, the *voltage* — the push behind the electricity — dips for a moment. Less voltage to your light bulbs means less brightness. Then, a heartbeat later, the motor is up to speed, the

current drops back to normal, the voltage recovers, and your bulbs are bright again.

That is the Blink. Inrush current → voltage sag → dim bulb → recovery. The whole thing happens in less than a second.

A modern building with thicker wires, a larger service panel, and a properly sized transformer on the pole outside will hide the Blink almost completely. You won't see it. The electrons will smooth it over before your eyes can catch it. But the physics is still there, every time the motor starts. The grid is just absorbing the punch quietly enough that you don't notice.

Now imagine a million air conditioners.

It is six o'clock on a Tuesday in August in a major city. The temperature has been climbing all day. Office workers are coming home. They walk in, drop their bags, and turn on their ACs. Not all at the exact same instant — but close. Within a few minutes, across a whole metropolitan area, millions of compressor motors are pulling their inrush currents nearly simultaneously.

Somewhere, far from any of them, power plants have to *catch* that. They have to push more current into the grid right now, this second, to keep the voltage from sagging citywide. If they can't, every bulb in the city would dip — and worse, sensitive equipment in hospitals, data centers, and factories would start to fault.

This is happening every minute of every day. The grid is one giant, synchronized machine, and somewhere — in control rooms in Pennsylvania and Texas and Ohio and California — there are people watching screens, watching frequency and voltage, making sure the grid catches every punch the world throws at it.

When the grid catches the punch, you don't notice. You make dinner. You watch TV. You go to sleep.

When the grid *misses* the punch, that is what 2003 was.

A tree branch in Ohio touched a sagging power line. A software bug in a control room meant no alarm went off. By the time anyone realized what was happening, the imbalance had cascaded across the entire eastern grid. Within seven minutes, the system had shed fifty million customers to save itself from physical destruction.

The Blink in my apartment was the grid catching a punch — barely, sloppily, but catching it. The Blackout was the grid missing one.

Here is the single most important fact about electricity, and it is so strange that most adults never really absorb it.

Electricity is made and used in the same instant.

The toast you ate this morning was generated in a power plant maybe a hundred miles away, less than one second before you ate it. The fan spinning in your room right now is being powered by a turbine spinning at this exact moment in a plant you have never seen, maybe burning gas or splitting atoms or being pushed by the wind. There is no warehouse of electricity sitting around waiting to be used. There is no giant bucket of stored electrons under the city. The grid is making your electricity *as you read this sentence*.

That is what makes electricity different from almost everything else humans deal with. Most things you can stockpile. You can buy a thousand cans of soup and put them in a basement and eat them next winter. You cannot do that with electricity. Batteries are a way around this — we'll spend a whole chapter on them later — but batteries are expensive and limited, and at the scale of a whole country they store only a tiny fraction of what we use.

This means the grid has to perform a feat of balance every single second of every single day. Supply has to match demand exactly. If a million people turn on their ACs at six PM, somebody, somewhere, has to be running a turbine harder right now to meet that load. If a

cloud passes over a giant solar farm, somebody has to make up the difference in the next few seconds, or the lights go dim.

The Blink in my apartment was that balance, almost failing, then recovering. The Blackout was that balance, failing, and the rest of the grid running for the exits.

Almost everything in this book — solar panels, battery walls, the grid of the future, why people in Texas lost power in a snowstorm, why China can build transmission lines that the United States can't — is downstream of this one fact. Electricity has to be made and used in the same instant. Everything else is a consequence.

If you remember nothing else from this book, remember the Blink.

Pay attention next time you turn on something big. A microwave. A hair dryer. A vacuum cleaner. Watch the lights — really watch them — and see if you can catch the dip. In a new building you probably won't. In an old building you might. Either way, you'll know what just happened. You'll know that the wires in your wall just experienced, in tiny form, the same thing the entire eastern seaboard experienced on August 14, 2003.

By the time we're done with this book, you'll see the grid everywhere. You'll understand why a tree branch in Ohio could darken Manhattan. You'll know why some roofs have solar panels and others don't. You'll know what a home battery actually does, and when it makes sense to buy one, and when it doesn't. You'll know what is happening when the lights in your school flicker during a thunderstorm. You'll know what the people in those control rooms are watching for, and why their job is one of the most quietly important in the country.

But first, the Blink. Watch for it tonight.



CHAPTER TWO

The Juggling Act



In a low building tucked behind a fence in Audubon, Pennsylvania, there is a room with no windows.

Inside the room, a wall of screens stretches from floor to ceiling. Lines of color trace across them, climbing and falling. Numbers blink. Maps of the eastern United States glow in different shades of green and yellow and the occasional red. There are desks in front of the wall, and at each desk a person sits in a swivel chair, watching, sometimes typing, sometimes picking up a phone.

This is one of the most important rooms in America, and almost no one outside the electricity industry knows it exists.

It is the control room of PJM Interconnection — the company that manages the flow of electricity for sixty-five million people across thirteen states and the District of Columbia, from northern Illinois to North Carolina. New York has its own grid operator. New England has another. California has another. Texas has its own. Together, a small number of these control rooms keep the lights on for the entire United States.

They never close. There is always someone in the chair. Day shift, swing shift, graveyard shift. Christmas morning, the middle of a

hurricane, the early hours of a new year. The grid never sleeps, so the operators never sleep.

What are they watching?

Mostly, they are watching one number.

The number is **60.000**.

It is the frequency of the electricity flowing through every wire in North America. The current in your wall is not flowing in one direction — it is *alternating*, swinging back and forth, sixty times a second. Every appliance you own is built to run on that exact rhythm. Plug a hair dryer designed for sixty cycles per second into a European outlet that runs at fifty, and the hair dryer will not work right. The motor will spin at the wrong speed. The clock will lose time.

That number — 60 hertz, or 60 Hz — is the heartbeat of the American grid. Every generator from Maine to Florida, and across most of Texas, is spinning in time with that beat. The huge steam turbines in coal and nuclear plants. The hydroelectric turbines under dams. The gas turbines that ramp up on hot afternoons. They are not just spinning *fast*. They are spinning *in lockstep*. All of them. At the same beat. Always.

If you could stand somewhere outside the planet and watch the eastern half of the United States as a single object, you would see thousands of giant rotating machines — turbines the size of trucks, generators the weight of houses — all moving together, as if one invisible hand were keeping time. That is what the grid is.

This is one of the most extraordinary engineering achievements in human history. And the room with no windows in Pennsylvania exists to keep it that way.

Here is why the number matters.

When demand exceeds supply — when more people are drawing electricity than the generators are producing — the generators *slow*

down. Not by much. A fraction of a percent. The number on the screens in Audubon drops to 59.98, then 59.97. The heartbeat is faltering.

When supply exceeds demand — when generators are producing more than people are using — the generators speed up. The number climbs to 60.02, then 60.04. The heartbeat is racing.

Frequency is the balance, made visible. It is the single most important measurement in the entire system, because it tells you, in real time, whether supply and demand are in balance. You don't have to track every house and every power plant separately. You just have to watch the frequency. If it's at 60.000, things are fine. If it's drifting, somebody somewhere is having a problem.

The operators in Audubon are watching that number to four decimal places. They can see it drifting before the public would ever notice. And they have tools to bring it back.

Picture a juggler. She has six balls in the air. If she throws them all up at the same height and catches them at the same rhythm, the act looks easy. But if one ball comes down a little fast, she has to throw it back up a little harder. If a ball lags behind, she has to slow her hand to meet it. Every ball, every second. Forever.

The grid operator is juggling supply and demand. The "balls" are power plants — coal plants, gas plants, nuclear plants, hydro dams, wind farms, solar farms, batteries. The operator does not control demand directly; that depends on whether millions of people are turning on their air conditioners or their coffeemakers or their TVs. But the operator can adjust supply.

If frequency starts to drop, the operator calls for more generation. Not by phone — by software, which sends signals to power plants telling them to spin a little harder, generate a little more. This adjustment is called **frequency regulation**, and it happens automatically thousands of times a day. A computer system called Automatic

Generation Control — AGC for short — handles the small adjustments in seconds. The operators handle the bigger ones, the weird ones, the ones the software is not sure about.

There is also something called **spinning reserve**. This is a power plant that is running, synced to the grid, burning a little fuel — but not actually generating much electricity. It is just *there*, hot and spinning, ready. The moment another power plant trips offline unexpectedly, the spinning reserve takes over within seconds. It is the grid's safety net, and it costs real money to maintain. A reserve plant is being paid to do mostly nothing except be ready.

This whole system — the regulation, the reserves, the operator watching the number — exists because of the single fact we ended Chapter 1 with: electricity is made and used in the same instant. There is no warehouse. So the warehouse has to be replaced by *vigilance*.

Here is the part that, when I first understood it, genuinely changed the way I think about the country I live in.

Every generator in the Eastern Interconnection — that's everything from Maine to Florida, west to the eastern Dakotas, and including parts of eastern Texas — is electromagnetically locked to every other generator. They are not just spinning at the same speed by coincidence. They are physically pulling each other along.

If one generator tries to slow down, the magnetic fields of all the others gently pull it back up to speed. If one tries to race ahead, the others gently hold it back. The whole thing is one giant machine, the size of half a continent, moving together. The biggest single machine humans have ever built.

There are three of these big machines in North America: the **Eastern Interconnection**, the **Western Interconnection** (everything west of the Rockies), and the **Texas Interconnection** (mostly the state of Texas, which famously chose to stay separate from the other two —

that's a story we'll come back to). The three machines are not synchronized to each other. They are connected by special direct-current links that let power flow between them, but their rotations are independent.

When you flip on a light switch in Philadelphia, you are participating in this machine. Your lamp pulls a tiny amount of current from the wires in your wall, which pull from the lines on the street, which pull from the substation, which pull from the transmission lines, which pull, finally, from generators hundreds of miles away that — at that exact moment — spin a fractionally tinier bit harder to feed your bulb. The whole machine flexes around you, infinitesimally, for as long as the lamp is on.

This is happening for every appliance in every house in every state, every second of every day.

For most of the twentieth century, the synchronized machine had a useful property: it was *heavy*.

Steam turbines and hydroelectric turbines have enormous rotating mass. They store kinetic energy the way a freight train stores momentum. If something on the grid suddenly changes — a power plant trips, a transmission line drops — the spinning mass of all those generators provides a few seconds of buffer. They slow down a tiny bit, releasing their stored energy into the grid, while the operators scramble to bring in replacements. This buffer is called **inertia**, and for a hundred years it was free. You got it just by having heavy spinning things.

Solar panels and battery storage do not have rotating mass. They use electronics to put electricity on the grid, and they can mimic the heartbeat — but they do not provide inertia for free. As more of the grid becomes solar and batteries, the system becomes less heavy. Less forgiving. Faster to react, and faster to fail.

This is one of the genuinely unsolved engineering problems of the energy transition. Engineers are working on something called **grid-forming inverters** — software-defined electronics that can fake inertia and even help stabilize a grid without any spinning mass at all. The technology is real and improving fast. But it is not yet fully deployed, and the grid is not yet ready to run on solar and batteries alone.

This is why a chapter on storage is coming later in this book. And why a chapter on the future is coming after that. The juggling act is not getting any easier.

In the control room in Audubon, the day shift ends in the late afternoon. The operator who has been watching the number for eight hours stands up, stretches, and walks to a meeting room with the operator coming on for the swing shift. They go over what happened — a generator that came offline this morning, a line that has been running hot, a forecast of thunderstorms over Maryland. They drink coffee. They sign a logbook. Then the new operator walks out to the chair, sits down, looks up at the wall of screens, and watches the number.

60.001.

60.000.

59.998.

60.000.

You are reading this sentence in a country where, right now, somebody you have never met is watching that number, making sure your lights stay on. They are good at their job. They have been doing it for years. Most of them love what they do, even though almost nobody outside the industry knows they exist.

They are the jugglers. They will be back in this book when we get to August 14, 2003 — because on that day, in a different control room in Ohio, the juggling act came briefly and catastrophically apart.

But for now, the heartbeat is steady.
60.000.



CHAPTER THREE

The Long Road



If you have ever driven on a highway in the United States, you have seen them.

Giant lattice-work towers, taller than most office buildings, marching in a long line across farmland and hills and forests. The wires between them sag in graceful curves — three of them, sometimes six, sometimes more. The towers keep going as far as you can see. They cross rivers. They climb ridges. They thread through gaps in the mountains.

Where do they come from? Where are they going?

The honest answer is that they are going from somewhere with a lot of electricity to somewhere with a lot of people, and almost everything else about the grid — the synchronized machine from the last chapter, the lights in your house, the air conditioner at your school — depends on them.

This is a chapter about wires. About why they have to be so big. About why they have to go so far. And about why building new ones may be the single hardest problem in clean energy today.

Power plants are not, as a rule, in cities. Cities have people, and people do not want a coal-burning plant or a nuclear reactor in their

backyard. Even cleaner plants are usually built far from where the electricity gets used.

There are also good *natural* reasons. The biggest hydroelectric dams need rivers in steep valleys. The best solar resources are in deserts. The best wind resources are on the Great Plains and offshore. None of those places have very many people.

So the American grid was built with a kind of permanent puzzle. The places that *make* electricity are not, mostly, the places that *use* it. The Hoover Dam sits on the Arizona-Nevada border, far from any major city. But it powers Los Angeles, almost three hundred miles to the west. The hydroelectric dams of the Columbia River sit in eastern Washington and Oregon. They power Seattle and Portland and, through one of the longest transmission lines in North America, Los Angeles again.

To bridge that gap, you need wires that can carry a lot of electricity, a very long way, without losing too much of it on the trip.

This turns out to be harder than it sounds.

Here is the problem.

When electricity flows through a wire, it meets a tiny amount of resistance. The wire heats up. That heat is energy that was supposed to reach your toaster but instead went into warming the air around the wire. It is lost. Gone. Wasted.

The longer the wire, the more heat. The more current flowing through, the more heat.

How much heat? The important part is this: **the heat lost goes up with the *square* of the current.** Double the current, four times the heat. Triple the current, nine times the heat. Push too much current down a wire and the losses become enormous — or the wire physically melts.

For a long time, this was the wall. You could only run electricity a few miles before so much of it was lost as heat that the whole enterprise stopped making sense.

The way around the wall is one of the most elegant tricks in all of engineering. It is so important that the entire modern world depends on it. And almost nobody outside electrical engineering ever thinks about it.

The trick is voltage.

The amount of *power* flowing in a wire is not just the current. It is the current multiplied by the voltage. **Power equals current times voltage.** $P = I \times V$.

This means there are two ways to send the same amount of power down a wire. You can have lots of current at low voltage. Or you can have a little current at high voltage. The power delivered is the same.

But the losses — the heat — depend only on the current. So if you keep the voltage low, you need a lot of current, and you lose a lot to heat. If you crank the voltage up, you only need a little current, and you lose almost nothing.

This is why transmission lines have such enormous voltages. The wires you see on those giant lattice towers might be running at 345,000 volts. Some run at 500,000 volts. The newest "ultra-high" lines run at 765,000 volts in the United States, and up to 1,100,000 volts in parts of China.

For comparison, the outlet in your wall runs at 120 volts.

The lines are held high in the air by giant porcelain insulators, well away from anything that could touch them — or anything they could touch. If you got too close to a transmission line, the voltage could leap through the air to reach you. That is how dangerous they are. That is also how efficient they are.

But there is a problem.

The appliances in your house cannot use 345,000 volts. They would melt. They are designed for 120 or 240 volts. So somewhere between the power plant and your toaster, the voltage has to come back down.

The piece of equipment that does this is called a **transformer**. You have probably seen them without knowing it. The big gray canisters on top of utility poles in suburban neighborhoods? Those are transformers. The fenced-off grids of equipment beside highways, with humming sounds and warning signs? Those are *substations* — full of transformers. The slightly larger gray boxes that sit on the ground in newer developments? Also transformers.

A transformer is, mechanically, simple. It is two coils of wire wrapped around the same iron core. Electricity flowing through one coil creates a magnetic field in the iron, and that magnetic field creates electricity in the other coil. If one coil has more loops than the other, the voltage changes when the electricity moves from one to the other.

More loops on the output side: voltage rises. This is called *stepping up*. Used at the power plant. Fewer loops on the output side: voltage falls. This is called *stepping down*. Used at the substation.

So the actual journey of electricity from the Hoover Dam to a lamp in Los Angeles goes something like this:

The generator produces electricity at maybe 20,000 volts. A transformer at the plant steps it up to 500,000 volts. The electricity races across the desert on transmission lines. A substation outside Los Angeles steps it down to 13,000 volts. Smaller transformers on neighborhood poles step it down again to 240 volts. The wire enters a house and powers the outlet, which delivers it to the lamp at 120 volts.

Voltage up, current low, losses low, distance covered. Voltage down at the destination, current up to a level appliances can use.

The whole grid, all the way from the dam to the lamp, is just this trick — performed thousands of times along the way.

In the 1880s, this was not yet a settled matter. There was a great public argument over how electricity should be delivered, and the two sides hated each other.

On one side was Thomas Edison, who had built the first commercial electricity system in lower Manhattan. Edison ran his system on **direct current**, or DC — electricity flowing steadily in one direction. The trouble was that the transformer trick we just learned does not work with direct current. With DC, you could not easily step the voltage up and down. So Edison's power plants had to be very close to their customers — within about a mile — and the wires carried thick, heavy currents at low voltage. To cover a city, you needed a power plant every few blocks.

On the other side was George Westinghouse, with help from a brilliant inventor named Nikola Tesla. Westinghouse and Tesla championed **alternating current**, or AC — electricity that swings back and forth, the way the grid still works today. Alternating current has a useful property: a transformer *does* work with it. So you could generate AC at one voltage, step it up for transmission, step it down for distribution. One power plant could light a whole city, with the plant located far away from anyone.

The argument got nasty. Edison's people staged public demonstrations electrocuting animals with alternating current to prove it was dangerous. Westinghouse and Tesla, for their part, pointed out that you could deliver electricity to far more people, at far lower cost, with AC.

In the end, AC won. The grid you live on today — the synchronized machine from Chapter 2, all sixty hertz of it — is Tesla's grid.

But here is the interesting twist.

In the last fifty years, direct current has quietly returned. Not for distribution. For the very, very long lines.

It turns out that for *enormous* distances — hundreds or thousands of miles — DC has some advantages AC does not. AC lines lose extra energy from a kind of pulsing magnetic effect along their length, which DC does not. DC lines can also carry more power on the same physical wire. And DC makes it easier to connect two electrical systems that aren't synchronized to each other.

So engineers built converters — massive electronic stations that can change AC to DC and back. The electricity flows out of the power plant as AC, gets converted to DC for the long haul, travels hundreds of miles as DC, then gets converted back to AC at the destination, where the local grid still runs on Tesla's old design.

These are called **high-voltage direct current** lines, or HVDC. The Pacific DC Intertie, which runs almost 850 miles from the Columbia River in Oregon to Los Angeles, is one of them. It was completed in 1970 and is still working. China has built the longest in the world: a single line running more than 2,000 miles at over a million volts, from hydroelectric dams in the western mountains to the cities on the eastern coast.

Edison was right about direct current, eventually. He was just about a hundred years too early.

If transmission is so clever, and the physics is so well understood, why don't we just build more lines? Why is the United States still struggling to connect its wind farms in Wyoming and its solar farms in Arizona to its cities on the coasts?

The answer is not engineering. The answer is permitting.

Building a new high-voltage transmission line in the United States requires getting permission from every single landowner whose property the line crosses. It also requires approval from every state the line passes through. Each state has its own rules. Each county

has its own rules. Each landowner has their own opinion. Every step can be appealed in court.

A famous example: a project called Northern Pass was supposed to bring clean hydroelectric power from Quebec into New Hampshire and from there into the rest of New England. It would have lowered electricity prices, reduced emissions, and increased reliability. It was proposed in 2010. It was killed in 2018 by opposition from New Hampshire residents who did not want transmission towers on their land. New England is still paying more for dirtier electricity than it would have with Northern Pass.

A more recent project, SunZia, brings wind power from New Mexico to Arizona and California across about five hundred miles. It was first proposed in 2006. It took seventeen years of permitting fights, environmental reviews, and lawsuits before construction began. Wind power was a relatively young technology when the project started; by the time it was approved, wind had become one of the cheapest electricity sources in the world.

This is why, in earlier chapters, I have mentioned permitting paralysis as a real American problem. It is the bottleneck for almost everything in clean energy. You can have the best wind and solar potential in the world. You can have the cheapest hardware. None of it matters if you cannot get the electricity from where it is generated to where people live.

By comparison, China can build a 2,000-mile transmission line in about three or four years. We will come back to that comparison — and what it costs underneath — in a later chapter.

For all the cleverness, the modern American grid still loses about **five percent** of all the electricity it generates to transmission and distribution losses along the way. Five percent does not sound like much. But the United States generates roughly four trillion kilowatt-hours per year. Five percent of that is two hundred billion kilowatt-

hours — more than the entire annual electricity use of many countries.

That is what is leaking out of the wires as heat, between the power plant and your outlet, every year. It is the price the grid pays for distance.

In a world that needs more clean energy fast, in places far from where the electricity is wanted, the long road is not getting shorter. It is getting longer. And the wires we are going to need over the next fifty years are wires that have not yet been built, on land that has not yet been crossed, through political and legal fights that have not yet been won.

The next time you drive past one of those lattice towers, look up. They are not background scenery. They are the answer to the question of how a power plant in one state lights a lamp in another. They are the long road.

We just have to build a lot more of them.



CHAPTER FOUR

Seven Minutes in Ohio



Tuesday, August 14, 2003. Akron, Ohio. The FirstEnergy control room. It is the early afternoon of an ordinary hot summer day, and four men are sitting at desks looking at screens.

Outside, the temperature is in the upper eighties. Air conditioners across the eastern United States are humming. Demand on the grid is high but manageable — well within what the system was designed to handle. The screens in the Akron control room show what they always show: maps of the transmission network, line loadings, voltage and frequency, alerts. The frequency is steady at 60.0. The lines are loaded but not overloaded. Nothing is flashing red.

The men have done this job for years. None of them is panicking. None of them has any reason to.

Within two hours, this control room will become the most consequential place in North America. The decisions made — and not made — in this room will plunge fifty million people across eight U.S. states and one Canadian province into darkness.

And none of the men in the room will see it coming.

To understand what went wrong, you have to understand what an alarm system in a control room is for.

The operators are watching, yes. But they cannot watch every single piece of equipment every second. The grid has thousands of components. So a piece of software — in FirstEnergy's case, a system called XA/21, built by General Electric — runs constantly in the background, comparing every measurement against what is expected, and *alerting* the operators when something is off. A line gets overloaded: alarm. A voltage drops below normal: alarm. A generator goes offline: alarm. The screens flash. A sound plays. The operators look up and respond.

Without the alarm system, the operators are mostly blind. They are watching perhaps a dozen screens out of hundreds, and trusting the software to tell them which screens to watch.

At 2:14 PM on August 14, the XA/21 system in Akron silently crashed.

A bug — what programmers call a "race condition," where two parts of the software tried to access the same data at the same instant and tangled each other up — caused the alarm system to stop functioning. It did not beep. It did not say *I am no longer working*. It just stopped. The operators in the control room had no idea anything was wrong.

The screens still updated. The numbers still scrolled. From the operators' point of view, everything looked exactly as it had a minute earlier. Quiet, normal, manageable.

This is one of the most dangerous failure modes any system can have. The thing that is supposed to warn you when something is wrong is itself wrong, and it does not tell you.

At 3:05 PM, fifty-one minutes after the alarm system died, a high-voltage transmission line called Harding-Chamberlin started to sag.

The day was hot. The line had been carrying a lot of current. From Chapter 3, you remember that current heats wires. Hot wires expand. Expanding wires sag lower toward the ground. Most of the

time this is fine, because the lines have been designed with sag built into the calculations and the right-of-way underneath them is supposed to be clear of tall things.

But somewhere along the Harding-Chamberlin line, a tree had grown taller than it was supposed to. FirstEnergy's vegetation management program — the people who trim trees in transmission corridors — had not gotten to that section recently enough. The hot, sagging wire touched the tree.

When a high-voltage line touches a tree, it does not gently spark. It arcs. The electricity flashes from the wire through the tree to the ground in a brilliant burst. The line's protection equipment — designed to prevent fires and equipment damage — instantly disconnects the line from the rest of the grid.

The Harding-Chamberlin line went dark.

In the Akron control room, no alarm sounded.

Here is the dangerous thing that happens when a line drops.

The current that *was* flowing through the Harding-Chamberlin line does not just disappear. The electricity still has to get from where it is being generated to where it is being used. It just has to find a different path. So it redistributes itself across the remaining lines in the area, raising their loadings.

A few of those neighboring lines are now carrying more current than they were before. More current means more heat. More heat means more sag.

At 3:32 PM, twenty-seven minutes after the first failure, the Hanna-Juniper line sagged into a tree and tripped. Same physics. Same failure mode. Same silent control room.

Again, the current redistributed across the surviving lines. Again, those lines began to heat up and sag further.

At 3:41 PM, the Star-South Canton line tripped.

By now, phones in the FirstEnergy control room were starting to ring. The Midwest Independent System Operator — a regional grid operator that watches a wider picture than FirstEnergy does — called Akron to ask whether they were aware they had lost three major transmission lines in the last forty minutes.

The operators in Akron said they were not aware. They looked at their screens. The screens still showed everything fine. They began to suspect their system was lying to them.

But by then it was almost too late to matter.

Between 4:05 PM and 4:13 PM, the grid came apart.

Once enough transmission lines had failed in northern Ohio, the remaining lines could not carry the load. They tripped, one by one, like fuses blowing. Each trip pushed more current onto fewer lines, which then tripped, which pushed more current onto still fewer lines.

This is what engineers call a **cascading failure**. It is not a single event. It is a feedback loop that, once it starts, runs faster than humans can react.

At 4:06 PM, large parts of the grid in northern Ohio went dark.

At 4:10 PM, the cascade jumped across Lake Erie into Michigan and Ontario. It headed east.

At 4:11 PM, it reached western New York. The transmission lines feeding New York City began to overload as they tried to compensate for the failures upstream. The grid's automatic protection equipment, designed to prevent the entire system from suffering physical damage, started to disconnect huge regions from one another. Whole cities went offline in seconds.

At 4:13 PM, the cascade stopped. It stopped because there was nothing left to disconnect. Fifty million people across eight U.S. states and the Canadian province of Ontario were now without electricity.

In a subway tunnel under Forty-Second Street, a train rolled to a stop in the dark. Passengers looked up from their books. In an office tower in Manhattan, computer screens blanked at the same instant. In an apartment in Toronto, a refrigerator stopped humming. The lights at a stoplight in downtown Cleveland went out, and the car waiting at the red light slowly rolled forward into an empty intersection, not knowing what to do.

The juggling act, the synchronized machine, the heartbeat — all of it had failed in less time than it takes to listen to a song.

You do not get a fifty-million-person blackout without a *lot* of small things being wrong at once.

The official report on the 2003 blackout, written by a joint U.S.-Canadian task force, listed dozens of contributing causes. They are worth pausing on, because together they tell a story that applies to almost any large disaster.

A software bug. The XA/21 alarm system had a defect that GE did not know about. After the blackout, GE issued a fix, and similar systems around the world were updated.

Tree management. FirstEnergy had fallen behind on trimming trees in some of its transmission corridors. After the blackout, federal regulators imposed new mandatory rules — with fines — for vegetation management. Today, U.S. utilities trim trees on a strict schedule.

Lack of situational awareness. The Midwest Independent System Operator had the data to see the cascade coming, but its monitoring systems did not have a way to clearly display the developing problem across multiple utilities' territories. After the blackout, regional grid operators got new tools to see the bigger picture.

Aging infrastructure. Many transmission lines in the affected region were running closer to their physical limits than designers

had assumed. After the blackout, more line upgrades and new construction were prioritized.

No mandatory reliability rules. Before 2003, reliability standards for the U.S. grid were *voluntary*. After 2003, Congress passed legislation creating mandatory standards enforced by an organization called the North American Electric Reliability Corporation, or NERC. Utilities that violate the rules can now be fined millions of dollars per day per violation.

The blackout was not caused by any one person making a stupid choice. It was caused by many small problems lining up at once, and by a single bug that hid the unfolding disaster from the people who could have stopped it. The operators in Akron were not bad operators. They were good operators looking at lying screens.

This is the most important takeaway from any large engineering failure: **disasters rarely happen because someone did something wrong. They usually happen because many small things were wrong at once, and nobody could see the whole picture.**

The 2003 blackout caused somewhere between four and ten billion dollars in damage. Spoiled food in refrigerators. Lost wages. Industrial processes interrupted. Hospitals running on generators that were not designed for sustained use. Cell towers losing battery backup. Water pumps stopping — some communities lost water pressure entirely.

It killed people, too. Most blackout deaths are indirect — people who depended on medical equipment, people trapped in unventilated heat, traffic accidents at dark intersections. The official death toll from the 2003 blackout is around one hundred people. The unofficial figure may be higher.

Power was restored to most areas within twenty-nine hours. Some places were dark for several days.

For many people who lived through it, the lasting memory is not the inconvenience. It is the strange, peaceful experience of an enormous modern city suddenly hearing no air conditioners. People in New York that night went up onto rooftops and saw, for the first time in their lives, the stars over Manhattan.

After 2003, the United States grid became more reliable than it had been before. New rules, new software, new monitoring, new tree trimming, new mandatory standards. Another cascade on the scale of 2003 has not happened on the eastern grid since.

But the grid is still vulnerable. The basic facts have not changed: electricity has to be made and used in the same instant, the system is one giant synchronized machine, and feedback loops can still outrun human reaction time. New kinds of failure are becoming possible — failures that come from cyberattacks rather than software bugs, from extreme weather rather than overgrown trees, from the heat of a changing climate stressing equipment that was designed for a cooler world.

Texas would discover one of these new failure modes in February 2021, in a way that killed more people in a single week than the 2003 blackout did. That story is coming, four chapters from now.

For now, look up the next time you walk past a transmission tower. The lines are humming. The grid is balanced. The heartbeat is steady. Somewhere, in a control room you will never see, a person you will never meet is watching the screens.

They are watching, in part, because of what happened in Akron, Ohio, in the early afternoon of a Tuesday in August 2003.



CHAPTER FIVE

The Last Hundred Feet



A few years after I lived in the old Hoboken apartment with the blinking bulbs, I moved.

The new place was only a few blocks away. Same neighborhood. Same utility company. Same grid feeding the same streets. The only thing that changed was the building. The new one was newer, larger, built in this century rather than the last one. The wiring inside the walls had been put in by people who knew that twenty-first-century apartments would have central air conditioning, microwaves, dishwashers, gaming computers, and an EV charger or two somewhere in the future.

I turned on the AC the first hot night. The bulbs did not blink.

I stood there for a moment in my new living room and looked at the lamp. I turned the AC off. I turned it back on. The motor hummed to life. The bulbs stayed bright. Steady. Quiet. Boring.

That was when I understood something that had been hiding in plain sight for years.

The grid above my head had been the same grid the whole time.

The thing that had changed was the building.

The Blink from Chapter 1 was not really a grid problem. It was a building problem.

This is one of the most important distinctions in the entire book, and almost nobody outside the electrical trades makes it cleanly. So let's make it cleanly here.

When you flip a switch in your house, the electricity reaching your bulb has been on a journey through *two different systems*, run by different companies, regulated by different rules, built to different standards.

The first system is the **grid** — the synchronized machine, the transmission lines, the substations, the high-voltage wires. This is what we have been talking about for four chapters. It is the system run by your utility company and watched over by control rooms in places like Audubon, Pennsylvania.

The second system is the **building** you live in. The wires running through the walls. The service panel — sometimes called the breaker box — in the basement or the hallway closet. The outlets. The light switches.

These two systems meet at a place called the **service drop** — the wires that come down from the utility pole into your house, or up out of the ground in newer developments. Everything before the service drop is the grid's responsibility. Everything after it is the building's.

When the grid fails, you get a blackout. Everyone on your block goes dark. The utility sends a crew.

When the building fails, you get a blink. A flicker. A breaker that keeps tripping when you run the microwave and the toaster at once. A burning smell from an outlet that you should *absolutely not* ignore. The utility does not help with this. You call an electrician.

Most people never have to think about which kind of problem they are having. The reason most Americans never have to think

about it is that, for the second system to work reliably, somebody had to build the house correctly. And for old buildings, "correctly" was defined a long time ago.

There are about a hundred feet of wire between the transformer on the pole outside your house and the outlet next to your bed. Maybe a little more. Maybe a little less. Inside that hundred feet lives most of the small electricity problems most people will ever experience.

There are three usual suspects.

Undersized service. When your house was built, the electric company brought in a wire sized for the demand expected at the time. In the 1950s, that might have been 60 amps. In the 1970s, 100 amps. New houses today usually get 200 amps. A 60-amp service was fine when "high demand" meant a refrigerator, a few lights, and an electric stove. It is not fine when you add central air conditioning, a clothes dryer, a microwave, a dishwasher, a hair dryer, three televisions, a gaming computer, and a car charger. The wires coming in are too thin. The breaker box is too small. Every time you ask for a lot of current at once, the system strains.

Old wiring inside the walls. The actual wires running between your outlets and your panel might be older than you are. In houses built before the 1960s, the insulation around the wires is sometimes a fabric-and-rubber compound that becomes brittle with age. Touching a brittle wire can crack the insulation. Cracked insulation can cause short circuits and fires. There is also an infamous period in the 1960s and 1970s when American builders used **aluminum wiring** instead of copper, because copper had gotten expensive. Aluminum wiring is not inherently dangerous, but it expands and contracts more than copper, and the connections at outlets and switches can loosen over decades. Loose connections produce heat. Heat produces fires. The U.S. Consumer Product Safety Commission still

considers homes with the original 1960s-70s aluminum wiring to be a real fire risk.

A tired transformer outside. The gray canister on the pole — or the green box in the yard, in newer developments — was originally sized for the load of the buildings it serves. If those buildings were sized for 1965 demand, and the neighborhood has not been upgraded, the transformer is doing more work than it was designed for. It may run hot. It may sag in voltage when the neighborhood collectively turns on its air conditioners at 6 PM. The blink in your apartment may not actually be coming from inside your apartment at all. It may be coming from a transformer that has been on the pole since the Johnson administration.

The hot summer night in my old Hoboken apartment, the blinking bulb might have been any of these three. Probably some combination. The new building I moved to had been wired in this century by people who expected modern loads, was connected to a properly sized transformer, and had a panel large enough to absorb the inrush current of an air conditioner without flinching.

The grid was the same. The last hundred feet had changed.

Take a walk around an older American neighborhood. Look up.

You will see gray canisters mounted on the utility poles, every few houses or so. They are about the size of a small barrel. They hum slightly. They are usually painted a dull gray and have warning labels with lightning bolts on them. Sometimes the wires coming off them have insulation that is a little faded. Sometimes there is a leak of oily liquid on the side, which is the cooling oil that keeps the transformer from overheating. Sometimes they are covered in pigeon mess.

Almost nobody notices them.

These are the workhorses of the last hundred feet. Each one is a step-down transformer — the same kind of equipment we met in

Chapter 3, only much smaller. The wires coming down from the top carry the distribution voltage — usually around 7,000 volts, sometimes 13,000 volts. The wires coming out the bottom carry the much lower 240 volts that goes into your house.

A pole-mounted transformer typically serves anywhere from one house to a dozen, depending on the neighborhood. They are designed to last twenty to thirty years. Many of them have been hanging on those poles for forty or fifty.

When a pole transformer fails outright, the houses it serves go dark and the utility comes to replace it within hours. But long before they fail, transformers can age in ways nobody notices. They can lose efficiency. They can become unable to handle peak loads as well as they could when they were younger. They can be the silent reason a whole block has blinky lights every August.

Most of the time, the utility does not replace a pole transformer until it fails. Why? Because there are tens of millions of them in the United States, and replacing them all proactively would cost a staggering amount of money — money that would come from the customers in higher bills. It is cheaper to wait for them to fail and then fix them.

This is a choice. It is not a bad choice. It is a trade-off — the same kind of trade-off we will see again and again in this book. Reliable, affordable, clean: pick two.

The last hundred feet of every American house is about to get a lot of attention.

Here is why. For most of the twentieth century, the electrical demands of an average house grew slowly. A house built in 1950 used much less electricity than one built today, but the growth was steady and predictable. Utilities and building codes kept up.

Now the demands are growing faster. Here are a few of the things many American houses are getting, or about to get:

Heat pumps. Replacing a gas furnace and a separate air conditioner with one electric system. Heat pumps are three to four times more efficient than gas, but they need a much larger electrical circuit than the air conditioner alone. A heat pump can pull 40 amps continuously on a cold day.

Electric vehicles. A modern home EV charger draws 32 to 48 amps. That is roughly the same as running an electric oven full blast for as many hours as you charge.

Induction stoves. Replacing gas stoves with induction is faster, cleaner, and safer — but induction stoves can pull 40 amps too.

Heat pump water heaters. Replacing a gas water heater with an electric one. Another 30 amps when it cycles on.

If a house adds a heat pump, an EV charger, an induction stove, and a heat pump water heater — which is the standard package for fully electrifying a home — the total demand can easily exceed the capacity of an older 100-amp service. The grid above the house is fine. But the wires coming *into* the house, and the panel inside the house, were not designed for this much electricity.

In many American cities, getting a service upgrade — replacing your old panel, and possibly running a thicker wire from the pole — can cost five thousand dollars or more and take six months to schedule. There is a national shortage of qualified electricians. And in many neighborhoods, even if every individual house could be upgraded, the local distribution transformer would have to be upgraded too, because the combined new demand of all the upgraded houses would exceed what the transformer was sized for.

This is the bottleneck for electrification in the United States. Not the grid. The last hundred feet.

Years ago, I lived in Shanghai for nine years and never experienced a blackout. The grid was steady. The lights never blinked. The

country had built one of the most reliable urban grid systems in the world.

And the building I lived in was old — out of code, the wiring caught fire once, the handyman pulled charred copper out of the plaster while I watched.

The grid was fine. The building was a fire hazard. They were not the same system. They had not been built to the same standards. They had not aged at the same rate.

Most people, in most places, do not have these two layers in their heads as separate things. We say "the electricity is unreliable here" or "the electricity is great there" and we mean one thing. But the experience of electricity in a building is the product of two layers, multiplied together. A great grid feeding a bad building gives you Shanghai's burned wires. A great grid feeding a good building gives you boring, reliable, unremarkable electricity — which is the goal, by the way, even though "boring" sounds like a backhanded compliment. An okay grid feeding an old American building gives you the Blink.

You can fix one layer without fixing the other. You can also leave both broken. Which combination you live in is mostly about where you happen to be — and, within a given place, mostly about the building.

The last hundred feet is where most personal decisions about electricity live.

Can you install solar on your roof? That depends on your panel, your wiring, your local transformer, and your utility's rules. We will get to that.

Can you afford a home battery, and when does it make sense? That depends on your last hundred feet too.

Should you electrify everything in your house, and in what order? That depends on your existing panel, your wiring, and your utility's incentive programs.

For now, look around your house. Find the breaker box. The panel will have a number on it — 60, 100, 150, or 200 amps. That number is how much electricity your last hundred feet was built to deliver, on the day someone installed it.

Look at the pole transformer on the street, if your neighborhood has them above ground. If it is dented, leaking, or covered in pigeons, it has been there a while.

Look at the outlets. If they are the old two-prong style rather than three-prong, your wiring is probably older than the moon landing.

None of this means your house is dangerous. Old houses in good repair can be perfectly safe. But it is good to know what you are working with.

The Blink, in retrospect, was a small message from the system. It was telling me that the last hundred feet had been built a long time ago, for a different kind of life, and was now being asked to do more than it had ever been designed to do.

It is, more and more, what the entire country's last hundred feet is being asked to do.



CHAPTER SIX

The Sun Doesn't Ask Permission



Imagine a spring afternoon in California. The sky is clear. The temperature is in the seventies. People are at work or school. Air conditioners are not running because it is not yet hot enough. Lights are off in most homes.

And across the deserts and rooftops of the state, several million solar panels are producing electricity at full output.

There is so much solar electricity flowing onto the grid right now that the operators at California's grid control center — CAISO, the California Independent System Operator — have a problem most engineers in the twentieth century would have considered absurd. There is more electricity than anyone in the state can use.

So the operators do something genuinely strange. They start paying their neighbors — Arizona, Nevada, sometimes states even farther away — to *take* the excess electricity off their hands. Sometimes the wholesale price of electricity, the price that big buyers pay, drops below zero. A generator selling electricity at *minus* twenty dollars per megawatt-hour is actually paying customers to take it.

When even paying neighbors is not enough, the operators tell solar farms to **curtail** — to disconnect from the grid and stop pro-

ducing. Beautifully working solar panels, sitting in the sun on a clear day, are deliberately turned off. The electricity they could have made is simply not made. Free, clean energy is allowed to bounce off the panels and warm the desert air instead of becoming usable power.

In 2024, California curtailed 3.4 million megawatt-hours of solar and wind energy this way — a 29 percent increase from the year before. Solar accounted for 93 percent of it. That is enough electricity to power roughly 300,000 American homes for a year, simply not generated.

If this sounds wasteful and strange to you, you are paying attention. The reason it happens is not a flaw of the engineers. It is a flaw of the *timing* — and the timing is one of the deepest, most genuine challenges of the energy transition.

For the entire history of electricity, generation followed demand.

If people wanted more electricity at six in the evening — and they did, predictably, every day, because that is when they came home and turned on their lights and started cooking — then the grid operators ramped up the power plants to meet that demand. Coal plants, gas plants, hydroelectric dams, nuclear reactors. These were all what engineers call **dispatchable** sources. You could tell them when to produce more electricity and they would produce more electricity. The operator was in charge of the timing. The plants did what they were asked.

Demand goes up: ramp generators up. Demand goes down: ramp generators down.

The system was simple enough that it could be managed by humans in control rooms with telephones, which is exactly how it was managed for most of the twentieth century.

Solar and wind do not work this way.

Solar panels generate electricity when the sun is shining. The brighter, the more. They produce nothing at night, less in the morn-

ing and evening, less on cloudy days, almost nothing in a storm. They do not care whether the grid needs the electricity right now.

Wind turbines generate electricity when the wind is blowing. They do not care about your dinner.

This is what engineers mean when they call these sources **intermittent** or **variable**. The sun does not ask permission. The wind does not check the schedule.

If you want a grid that runs on solar and wind, you have to flip the entire hundred-year-old logic of electricity inside out. Instead of generation following demand, demand has to follow generation — or, more precisely, you have to find ways to *store* generation until the time when demand needs it.

This is a much harder problem than it sounds.

To see exactly how hard, take a look at something the people who run California's grid call the **duck curve**.

The duck curve is a chart. It shows electricity demand on the California grid over a single day, from midnight to midnight, *after* you subtract the electricity being produced by solar panels. This is called the **net load** — the amount of electricity the grid operators have to produce from other sources to meet what people are actually using.

In 2014, before California had much solar, this chart looked like a smooth, gentle hump. Demand was lowest in the early morning hours. It rose through the morning, plateaued in the afternoon, peaked around six or seven in the evening when people came home, and fell back down at night. A normal-looking curve.

Today, the chart looks like a duck.

Here is why. In the middle of the day, when solar panels are producing maximum output, the net load — the demand the grid still has to meet from other sources — drops dramatically. The chart sags downward into a deep belly. Sometimes the belly goes below the morning floor. There is less demand for non-solar electricity at noon

than there is at four in the morning. (In CAISO today, solar supplies almost *half* of all electricity demand between 8 AM and 4 PM. The midday belly is that big.)

Then the sun starts to set. Solar production falls off a cliff. People come home from work and turn on their air conditioners, ovens, lights, televisions. The net load shoots upward, almost vertically. The chart rises into a sharp peak. That is the head of the duck.

This is what the duck curve looks like. A flat morning back. A sagging midday belly. A sharp evening neck. A pointed sunset head.

The duck is not a small problem. Between roughly 4 PM and 7 PM on a spring evening in California, the grid has to *ramp up* about 13,000 megawatts of generation — the equivalent of about thirteen large nuclear power plants worth of output — in three hours. Every day. As the sun sets.

You cannot ramp a nuclear plant up that fast. You cannot ramp a coal plant up that fast. Even gas plants, which are the fastest-ramping of the conventional sources, strain at this. The grid has to keep enough natural gas plants warm and ready, all day long, to come online when the duck's neck rises.

The duck, in other words, is the shape of what intermittency does to a grid that still has to keep the lights on.

There is another problem hidden in the duck. The midday belly — the part where solar is producing more than the state needs — is the source of the strangeness we started this chapter with.

When solar is over-producing, the electricity has to go *somewhere*. Electricity cannot pile up. You cannot stockpile it in a warehouse. The grid balance from Chapter 2 requires that every kilowatt being produced is being used in the same instant.

So California has three choices when its solar floods the grid.

Send it somewhere else. Sell it to Arizona, Nevada, even down to Mexico if the transmission lines exist. Sometimes the neighbors do

not need it either. Then the price goes negative — California pays Arizona to take it. This is genuinely happening on real grids in the United States right now.

Use it for something flexible. Charge batteries. Pump water uphill into a reservoir. Run industrial processes that can shift their schedule, like aluminum smelting or hydrogen production. The trouble is that there are not yet enough of these flexible uses to absorb the excess. There will be more in the future. Right now, not enough.

Throw it away. Tell solar farms to curtail. Tell wind farms to feather their blades and stop spinning. The free electricity is not generated.

Curtailment is the option of last resort. The energy is wasted. The owners of the solar farm lose revenue. Nothing useful happens with the photons that hit those panels. But the grid stays balanced, because that is the iron law.

It is not just California. Texas curtails wind regularly. Hawaii curtails solar. Germany has paid surrounding countries to take excess wind power during stormy nights when the wind is howling and demand is low.

This is a strange thing to read in a book that is, in part, about cleaning up our grid. We have an abundance of free, clean electricity at certain hours — and we are throwing some of it away. The reason is not that anyone wants to. The reason is that we have not yet built enough ways to *catch* it.

Wind is intermittent in different ways than solar is, and the ways are sometimes worse.

The most awkward fact about wind, especially in places like the central United States, is that it often blows hardest at night and dies down during the day. The wind farms of Texas, Oklahoma, and Iowa generate most of their electricity between midnight and dawn

— exactly when nobody wants it. Then the wind slackens during the day, when demand is high. Then it picks up again at night, when demand falls.

This is the opposite of helpful.

Wind has another, more dangerous habit. During cold weather, high-pressure systems sometimes settle over large regions and stay there for days at a time. The air is still, the sky is clear, the temperature is freezing. There is no wind. Across hundreds of miles, wind farms stop turning, all at once, for an entire week.

This is the worst possible time for wind to fail, because cold weather is when people most need electricity for heat. A grid that has come to rely on wind, but does not have a backup plan for the windless weeks, can find itself in a very bad place.

Texas found itself in a very bad place in February 2021. That story is coming.

When you read in the newspaper that a country has installed "300 gigawatts of solar," you might assume that 300 gigawatts is, well, 300 gigawatts. It is not.

A solar farm rated at 100 megawatts produces 100 megawatts only at the brightest moment of the brightest day of the year, with the panels at perfect temperature and perfectly clean. Over the course of a whole year, the average output of that solar farm is closer to 25 megawatts. The ratio of average to maximum is called the **capacity factor**. For solar in most of the United States, it is around 25 percent.

For wind, the capacity factor is around 35 to 40 percent on good land, and as high as 50 percent at sea.

For a coal plant, the capacity factor used to be 70 percent or more, though it has fallen in recent years as cheaper sources push coal off the grid.

For a nuclear plant, the capacity factor is over 90 percent. Nuclear plants run almost all the time. They take a few weeks off every year and a half for refueling, and that is about it.

This means when you compare a "1,000-megawatt coal plant" with "1,000 megawatts of solar panels," you are not comparing equivalent things. The coal plant will produce something like three to four times as much electricity over a year as the solar will, even if the nameplate ratings are equal. To replace a 1,000-megawatt coal plant with solar, you need to install three to four times as much solar — *and* you need to figure out what to do for the nighttime hours.

This does not make solar bad. Solar is now the cheapest source of new electricity in most of the world. But the comparison "we installed enough solar to replace a coal plant" is almost always misleading. The math is not that simple.

Engineers and policymakers have been working on the intermittency problem for thirty years, and they have a partial toolkit.

Geographic diversity. The wind may not be blowing in Iowa, but it might be blowing in Wyoming, or off the coast of Massachusetts. The sun may not be shining in Seattle, but it is shining in Arizona. If you can move electricity across long distances, you can smooth out a lot of the variability. This is one reason new transmission lines — the long road from Chapter 3 — are so important to the energy transition. It is also one reason permitting paralysis is a real cost.

Demand response. Instead of always shifting generation to meet demand, the grid is starting to shift *demand* to meet generation. Pay electric car owners to charge at noon, when solar is abundant, instead of at 6 PM. Set water heaters and washing machines to run when the grid is flush. Pay big industrial users to ramp down for an hour when demand spikes. This is the smart grid we will spend a whole chapter on later in this book.

Overbuild and curtail. Build so much solar and wind that even on a bad day, there is still enough. Accept that on good days, you will be throwing energy away. The economics of this strategy are actually getting reasonable, because solar and wind are now so cheap that overbuilding is not as expensive as it sounds.

Storage. Catch the excess midday solar in batteries. Pump water uphill in reservoirs. Compress air into caverns. Heat blocks of sand or molten salt. Then release the stored energy in the evening when the duck's neck rises.

Storage is the big one. It is the reason the next chapter exists.

Solar and wind are now the cheapest sources of new electricity in many parts of the world. The price of solar panels has fallen by more than 90 percent in fifteen years. The price of wind turbines is a fraction of what it was. In sunny places, the wholesale cost of solar electricity is now lower than the wholesale cost from any other source — fossil or otherwise.

But "cheapest per kilowatt-hour at noon" is not the same as "cheapest grid." A real grid needs electricity at every hour of every day, in every kind of weather. The cost of the *system* — solar plus storage plus backup plus transmission plus the grid services that keep frequency stable — is higher than the cost of solar panels alone. How much higher depends on local geography, on existing infrastructure, on the mix of other sources available, on how much intermittency the grid is willing to absorb.

The honest summary, the one most cheerful headlines about clean energy will not give you, is this: **the cheapest electricity in the world is now intermittent, and learning to live with intermittent electricity is the central engineering project of the next two decades.**

This is a real, unsolved problem. Engineers are working on it every day. It is being solved in pieces, in different places, in different ways. It is not yet solved.

The duck curve, when you understand what it is showing, is not a sign of failure. It is a sign of success. The duck exists because California has so much solar that it has run out of demand to absorb it at midday. That is what winning at solar looks like.

California's solar and wind capacity grew from 9.7 gigawatts in 2014 to 28.2 gigawatts by the end of 2024 — almost tripling in a decade. The duck is the visible mark of that buildout. The duck is also the bill that comes with it.

The challenge now is what to do with the duck.

What to do with the midday belly, when there is more clean electricity than the grid can use. What to do with the evening neck, when the sun sets and the city needs power. What to do with all those hours in between, when supply and demand stop dancing with each other and start fighting.

The answer, as much as there is one, is storage. Catching the noon sun and releasing it at six. Catching the midnight wind and releasing it at three in the afternoon. Making the grid able to remember.

That is the next chapter.



CHAPTER SEVEN

Catching Lightning in a Bottle



Tucked into the Appalachian Mountains of western Virginia, near the West Virginia border, there is a small reservoir at the top of a mountain. There is another, larger reservoir at the bottom of the same mountain. Between them runs a tunnel, and inside the tunnel are six enormous water pumps, each as big as a house.

When electricity is cheap and abundant — at midnight, say, when nuclear plants are running steadily and nobody is awake to use the power — the pumps run. Water is lifted from the lower reservoir to the upper one. By morning, the upper reservoir is full.

When electricity is expensive — say, on a hot summer afternoon when air conditioners are humming and demand is peaking — gravity goes to work. Water is allowed to fall back down through the tunnel, spinning the pumps in reverse so that they now act as turbines. The turbines drive generators. The generators produce electricity.

This is the Bath County Pumped Storage Station, completed in 1985. It can store and release about 24,000 megawatt-hours of energy. It can produce 3,000 megawatts of power — about the same as

three large nuclear reactors — for almost a full day before the upper reservoir runs dry.

It is, by some measures, the largest battery on Earth.

It is not a battery. It is a mountain.

When you hear "energy storage," you probably picture batteries. The lithium-ion battery in your phone. The Tesla Powerwall on a neighbor's garage. A Tesla Megapack at a utility-scale battery farm out in the desert.

Batteries are important. Batteries are growing. But for most of the last hundred years, almost all of the energy storage on the world's electrical grids has been water in reservoirs, exactly like Bath County. Worldwide, pumped hydro still accounts for about 95 percent of all installed grid storage capacity.

The reason is simple. Pumped hydro is what engineers call a "low-tech, high-confidence" solution. The physics is gravity. The hardware is water, concrete, and turbines. We have known how to build these since the 1890s. They last fifty or eighty years with maintenance. They have no battery cells to replace, no rare metals to mine.

The problem with pumped hydro is geography. You need a mountain. You need two places to put reservoirs, one above the other, separated by enough vertical distance to store useful energy. There are only so many sites in the world that meet these requirements, and many of the best ones already have a power plant on them. You cannot just decide to build a Bath County in Kansas.

So the world has, in the last fifteen years, started building a different kind of storage. Not because batteries are better than mountains — they are not, for the things mountains can do. But because we are running out of mountains, and we need to store electricity everywhere, not just in the Appalachians.

In March 2017, the state of South Australia had a problem. After a series of blackouts caused by extreme weather and an aging power system, politicians and engineers were looking for ways to make the grid more resilient. Onto a social media argument about the situation came Elon Musk, the CEO of Tesla, who made a public bet on Twitter: Tesla would build the largest lithium-ion battery in the world for South Australia, and would do it in 100 days from the day the contract was signed — or it would be free.

The bet was real. The contract was signed. And Tesla built it in 63 days.

The **Hornsedale Power Reserve**, as it was named, is a battery farm in the South Australian outback. The original installation was 100 megawatts of power and 129 megawatt-hours of energy. (It has since been expanded.) Connected to a wind farm, it could store the wind farm's output when the wind was blowing too hard and discharge it back when the wind died.

In its first year of operation, Hornsdale was estimated to have saved South Australian electricity customers about fifty million dollars by reducing the cost of "grid services" — the small, fast adjustments that keep frequency stable. It also responded to several outages on the South Australian grid faster than any conventional power plant could have. When a coal plant 1,000 kilometers away tripped offline, Hornsdale was injecting power onto the grid within 140 milliseconds. Coal plants take minutes.

Hornsedale was the moment when grid-scale lithium storage stopped being a theory and became a real tool. After 2017, batteries started being built around the world at a pace nobody had anticipated. By the end of 2024, California alone had about 13 gigawatts of battery storage on its grid — more than any country on Earth had had ten years earlier.

The reason is partly that batteries got dramatically cheaper. The cost of a lithium-ion battery has fallen by about 90 percent in the last decade. Tesla's Megapack, the truck-sized battery that utilities now buy, costs roughly a tenth of what equivalent storage cost in 2010.

And the reason batteries are useful for grids is exactly the duck from Chapter 6. The midday belly fills the batteries with cheap solar. The evening neck draws the batteries down to meet demand. Batteries are how the grid catches the noon sun and releases it at dinner-time.

What about home batteries?

A Tesla Powerwall 3 — the home version of the same lithium-ion technology used in the Megapack — stores 13.5 kilowatt-hours of electricity. It costs somewhere between ten and fifteen thousand dollars installed, depending on your contractor. It is rated to last about fifteen years.

If you do the math, that works out to about ten to fifteen cents per kilowatt-hour cycled through it — on top of whatever the electricity cost to begin with. In most of the United States, retail electricity costs twelve to thirty cents per kilowatt-hour. So in many cases, the Powerwall does not pay back simply on the value of storing cheap electricity for later use. The savings from time-shifting the electricity are roughly canceled by the cost of the battery itself.

This sounds bad. It is actually a more nuanced story.

A home battery makes economic sense in most American homes today for two reasons that are not pure arbitrage.

Backup power. When the grid goes down — a storm, a wildfire-related power shutoff in California, a winter outage in Texas — the Powerwall keeps your fridge running, your phone charged, and (depending on size) your air conditioning going for some hours. For people who live in areas with frequent outages, or for people who

depend on medical equipment, this is worth more than the pure arbitrage math would suggest.

Time-of-use rates. In some states — especially California — your electric bill is no longer a flat rate per kilowatt-hour. It depends on the time of day. Peak rates between 4 PM and 9 PM can be three to four times higher than off-peak rates. Under those rate structures, charging the battery from cheap midday solar and discharging it during the expensive evening hours can produce real savings — enough to make the Powerwall worthwhile.

And from the previous chapter: whether the Powerwall is even *installable* in your house depends on your service panel, your wiring, and whether your utility allows it. New construction is easier. Old houses may need a panel upgrade first.

The honest summary, the one you will not always get from a Powerwall salesperson, is this: **a home battery is a hedge, not an investment.** It pays back if you value resilience and live somewhere with the right rate structure. It does not, in most places, pay back purely on the math of buying cheap and selling dear. The cost-benefit changes every year as battery prices fall and rate structures evolve, so the answer for your house may be different next year than it is today.

Lithium-ion is not the only kind of battery, and batteries are not the only kind of storage. Engineers are exploring a whole zoo of options, each with different trade-offs.

Flywheels. Big spinning wheels suspended in vacuum chambers. You spin them up with surplus electricity; you slow them down to recover the energy. Round-trip efficiency is high. Duration is very short — seconds to minutes. They are used for the fastest grid services, like keeping frequency stable. Remember the inertia problem from Chapter 2? Flywheels can help with that.

Compressed air. Use cheap electricity to compress air into underground caverns, often abandoned salt mines. Release the air later to spin turbines. The technology has been around since the 1970s. Round-trip efficiency is mediocre — maybe 50 to 70 percent. But the caverns are huge, so the storage durations can be very long.

Thermal storage. Convert excess electricity into heat — molten salt at 1,000 degrees, hot rocks, sand, even a giant tank of hot water — and reconvert it to electricity when needed. The efficiency depends a lot on what you are doing. If you can use the heat directly (for industrial processes or for heating buildings) instead of converting back to electricity, efficiency is excellent. If you have to convert it back, it is lower.

Gravity storage. Newer companies are experimenting with raising heavy weights up towers, then lowering them slowly to drive generators. This is the same principle as pumped hydro, but without needing a mountain or water. The technology is still being demonstrated; the economics are unclear.

Hydrogen. Use excess electricity to split water (H_2O) into hydrogen (H_2) and oxygen. Store the hydrogen. Burn it later in a gas turbine, or run it through a fuel cell. Hydrogen can be stored almost indefinitely, which is wonderful. But the round-trip efficiency is poor — only 30 to 40 percent of the original electricity comes back. Most of the energy is lost in the conversion both ways.

Iron-air batteries. A new kind of battery, very different from lithium. Iron-air cells use iron and oxygen — basically, rusting and unrusting iron on purpose — to store energy. They are slow. They are cheap. They have terrible energy density (you would never put one in a car). But they may be able to store energy for 100 hours at a tenth of the cost of lithium. The company Form Energy is building the first commercial iron-air battery installation in Minnesota.

Why so many approaches? Because no single kind of storage solves every problem. The grid needs storage at every time scale, from seconds (flywheels) to hours (lithium) to days (pumped hydro) to seasons (hydrogen, iron-air, maybe). The future grid will probably use a portfolio of these technologies, picking the right tool for each job.

If you remember one thing about storage, remember this:

Lithium batteries are great for four hours.

The standard size for a utility-scale lithium battery is something like 100 megawatts of power for four hours, which means 400 megawatt-hours of energy. After four hours, the battery is empty, and it takes a few hours to recharge.

Four hours is enough to handle a daily duck-curve cycle — fill the battery at noon, drain it during the evening peak. This is what California's 13 gigawatts of batteries are doing. They are flattening the duck.

But what happens when the wind does not blow for a week? What happens when a winter storm covers your solar panels in snow for days? What happens during a cloudy stretch in February with no wind?

You cannot solve a week-long shortage with a four-hour battery. Even if you had a thousand of them, you would drain them all on the first afternoon.

To run a grid on mostly solar and wind, you need something that can store energy for *days* or *weeks*, not just hours. This is the great unsolved problem of storage.

The candidates for long-duration storage are exactly the technologies described in the previous section that no one has yet deployed at large scale: hydrogen, iron-air, compressed air, thermal. Some of them will work. Some will turn out to be more expensive or more

limited than their promoters claim. We do not yet know which will dominate.

If you are a teenager reading this book and thinking about a career, this is one of the genuinely exciting open problems in the energy world. Whoever solves long-duration storage cheaply will probably make the energy transition possible, and might also become extraordinarily wealthy along the way.

A book about batteries has to be honest about one more thing: where the materials come from.

Lithium-ion batteries require lithium (mostly mined in Australia, Chile, and Argentina, with processing concentrated in China), cobalt (mostly mined in the Democratic Republic of the Congo, where some mines have been associated with serious labor and human rights abuses), nickel (Indonesia and the Philippines), and graphite (mostly China).

The supply chain is concentrated. China processes the majority of the world's battery materials, even when those materials come from somewhere else. The mining itself uses water, energy, and land. The labor practices in some mines have been documented as abusive, particularly in cobalt mining.

Newer battery chemistries are reducing or eliminating some of these problems. Lithium-iron-phosphate (LFP) batteries, which dominate the Chinese market and are spreading globally, do not use cobalt or nickel. Iron-air batteries, if they scale up, will use one of the most common materials on Earth. Sodium-ion batteries — which trade some performance for using one of the most abundant elements there is — are starting to appear in commercial products.

But it is fair to say that today's battery boom is, in part, built on mining. There is no version of "clean energy" that involves zero environmental impact. The honest framing is this: the climate damage from mining lithium is much smaller than the climate damage from

burning coal, and the human-rights problems can be reduced through standards and oversight — but the damage is not zero. It is something we are choosing to do because the alternative is worse.

This is a real trade-off, and you deserve to know about it. The world is not all good guys and bad guys. The world is people making choices about which problems to live with.

For a hundred years, the grid had almost no memory. Whatever was generated had to be used in the same instant. The grid was a system without a clock — it could only ever live in the present.

Storage gives the grid a clock. It gives the grid a kind of memory.

The noon sun in California can now be stored and released at dinner time. The midnight wind in Texas can be stored and released at noon. The duck is starting to flatten. The synchronized machine from Chapter 2 is learning to wait.

We are not there yet. Lithium gives us four hours. Pumped hydro gives us a day, but only in mountains. We do not yet have an economical way to store electricity for a week. The grids of 2050 will need that capability, and the engineers and entrepreneurs and policy people of the next twenty years are the ones who will have to invent it.

If you find yourself looking up at a Virginia mountain hillside with a quiet reservoir near the top, or driving past a fenced-off field of gray Tesla Megapacks humming in the desert heat, look at it for a moment. These are the experiments that may turn out to have built the world.

The next time the sun goes down behind your house, the grid still has to come up with the next few hours of electricity from somewhere. Increasingly, that somewhere is a place where, hours ago, the same sun put energy in.

The grid is learning to remember.



CHAPTER EIGHT

The Frozen Texan



Sometimes a disaster is in the forecast for days, and you can watch it coming.

In the first week of February 2021, meteorologists across North America began to notice something unusual happening in the upper atmosphere. The polar vortex — a swirl of frigid air that normally stays trapped around the Arctic — was wobbling. Pieces of it were breaking off and sliding south. By the second week of February, one of these pieces was heading directly into the central United States.

The weather service issued warnings. The temperature in Dallas would drop below zero for the first time in decades. In some parts of Texas, the temperature would fall by more than fifty degrees in twenty-four hours. The storm, eventually called Winter Storm Uri, would freeze a region that had not been seriously frozen since the 1950s.

People stocked groceries. People filled bathtubs with water in case the pipes froze. Schools and businesses closed. The state braced itself.

Then the storm arrived, and the grid started to fail.

By the time Texas thawed out a week later, somewhere between 246 and 700 people would be dead. It would be the worst weather disaster in Texas history. And almost none of those deaths were directly caused by the cold itself. They were caused by the failure of the grid.

Texas, you may remember from Chapter 2, has its own electrical grid.

When the great American grids were being interconnected in the twentieth century, most states gradually joined either the Eastern Interconnection or the Western Interconnection — the two giant synchronized machines that span the rest of North America. Texas chose differently. The Texas grid, run by an organization called ERCOT (the Electric Reliability Council of Texas), kept itself almost entirely separate from the others.

There was a reason for this. By staying separate, Texas's grid was not subject to federal regulation in the way that interstate grids are. The Federal Energy Regulatory Commission, or FERC, can set reliability rules for the Eastern and Western grids because they cross state lines. ERCOT crosses no state lines, and so has historically operated under Texas state rules instead — which were lighter and friendlier to producers.

This choice had real advantages. Texas built electricity infrastructure faster than almost any other state. Wholesale electricity prices were often lower than in California or the Northeast. Texas became, eventually, the largest wind-power producer in the country and one of the largest solar producers, partly because the regulatory environment made it easy for new generators to enter the market.

It also had a quieter cost, which would become very loud in February 2021.

After a smaller freeze in 2011 had caused brief blackouts, Texas's regulators had recommended that generators winterize their equip-

ment — wrap pipes, install heaters on critical components, protect against extreme cold. The recommendation was voluntary. Most generators did not do it, because doing it would have cost money, and the market did not reward them for being reliable in extreme weather. They were paid for electricity they actually delivered. If they were offline during a freeze, they simply did not collect that revenue. The math, for the generator, was to skip winterization and accept the occasional bad week.

The math, for the people of Texas, would turn out very differently.

The freeze began on Sunday, February 14, 2021 — Valentine's Day.

By Monday morning, demand on the Texas grid was setting all-time winter records. Millions of homes had electric heating systems — including many cheap electric resistance heaters that are extremely inefficient. People who normally used natural gas to heat their homes also needed electricity for blowers and pumps. Even gas customers were drawing more electricity than usual.

And at the same moment that demand was climbing, supply was falling apart.

Natural gas, which provided more than half of Texas's electricity, was failing in two ways. The wellheads in west Texas were freezing — water in the gas itself was forming ice plugs in the pipes. And the pipelines and compressor stations that moved the gas to power plants were freezing too. Some compressor stations had electric components that themselves needed electricity to run, which meant that as the grid lost generation, the gas system also lost ability to deliver fuel to the plants that still worked. It was a feedback loop in the worst possible direction.

Coal plants were failing as well — frozen pulverizers, frozen conveyor belts, frozen coal piles. A nuclear reactor at the South Texas

Project tripped offline because a water-level sensor froze (an embarrassingly small failure, given the size of the plant). Wind turbines, which had not been winterized for the Texas climate, iced over and stopped turning. Some solar panels were covered in snow.

By Monday morning, February 15, ERCOT's control room was watching its supply collapse minute by minute. The grid's frequency began to drop below 60 hertz — the heartbeat we met in Chapter 2. Then below 59.9. Then below 59.8.

If frequency stays below 59.4 hertz for more than nine minutes, automatic equipment begins disconnecting generators from the grid to protect them from physical damage. Once that starts, it cannot be easily stopped. The grid would collapse entirely.

At 1:51 AM on Monday morning, ERCOT made the only decision left: they began ordering utilities to cut power to their customers. The plan was to rotate the blackouts — to leave each neighborhood dark for thirty or forty-five minutes at a time, then move on. This was supposed to bring demand down to a level the surviving supply could meet.

The plan did not work as intended.

The problem with rolling blackouts is that they are surprisingly hard to actually roll.

The grid is not divided into neat blocks the size of city neighborhoods. It is divided into circuits that follow the historical accidents of how transmission lines were built, which substations serve which areas, which lines were upgraded in which decades. Some circuits cannot be disconnected without taking down a hospital, a fire station, a water treatment plant, a natural gas compressor station. Those circuits are protected — they have to stay on.

Once the protected circuits in each neighborhood are accounted for, what is left is often a much smaller set of circuits that can actually be turned off. And so the burden of rotating blackouts falls

disproportionately on those circuits — and on the people who happen to live on them.

In Texas, during the freeze, some neighborhoods stayed dark for fifteen minutes. Other neighborhoods, just a few blocks away, stayed dark for four days.

For people who lost power in subzero temperatures, the first few hours were uncomfortable. The next few hours were dangerous. By the second day, in homes that were not built to retain heat, the indoor temperature was approaching the outdoor temperature. Pipes were freezing and bursting, flooding kitchens with ice water. People wore every layer of clothing they owned and huddled in bathrooms with the doors closed.

Some people, desperate for warmth, brought charcoal grills inside. Charcoal produces carbon monoxide, which is colorless and odorless and lethal. Multiple families died this way, parents and children together in their living rooms.

Others ran cars in attached garages to warm up. Carbon monoxide killed them too.

Others left their homes to sit in their running cars in the driveway, just to charge their phones. Some of those people, exhausted from cold and sleeplessness, fell asleep with the engine running. They did not wake up.

In a Houston-area suburb, an eleven-year-old boy died in his bed of hypothermia during the night. His mother had been keeping him as warm as she could. The house had been without power for nearly two days.

He was not alone. Across Texas, that week, somewhere between 246 and 700 people died from causes directly traceable to the grid failure. The lower figure is the official state count. The higher figure comes from analyzing excess mortality — comparing how many people actually died that week to how many would normally have

died in a typical February week — and is what most independent researchers believe is closer to the truth.

After the disaster, ERCOT's CEO told a state legislative hearing that the Texas grid had come within four minutes and thirty-seven seconds of complete catastrophic failure.

If ERCOT's frantic measures had not worked — if they had not been able to shed enough load by cutting power to millions of homes — the grid's frequency would have fallen below the safety threshold, and the entire Texas Interconnection would have collapsed.

What does "complete collapse" mean? It means every generator disconnecting from the grid simultaneously. It means a state of thirty million people suddenly with zero electricity, *including the equipment used to start power plants*. Restarting a grid from a complete collapse is called a **black start**, and it is one of the most difficult operations in all of electrical engineering. It requires bringing generators online one by one, using small backup generators (themselves cold and possibly frozen) to power the equipment needed to start the bigger ones. Each generator, once started, has to be carefully synchronized to the small piece of grid being slowly rebuilt.

If the Texas grid had fully collapsed in February 2021, experts at the time estimated it could have taken *weeks to months* to restore power across the state. In subzero temperatures. With pipes already freezing. With food rotting in dark refrigerators.

That is what four minutes meant.

In the months after the freeze, Texas politicians held hearings. ERCOT's CEO was fired. Several of the state's Public Utility Commissioners were forced to resign. Investigations were launched.

In 2021 and 2022, the Texas legislature passed laws requiring power plants and natural gas facilities to winterize their equipment.

The Public Utility Commission established new rules. ERCOT changed how it operated.

When the next major winter storm hit Texas in December 2022, the grid mostly held. There were some outages, but not a collapse. The winterization rules had partly worked.

But several things did not change.

Texas remained part of its own interconnection, separate from the rest of the United States. Proposals to build major transmission ties with the Eastern Interconnection — which would have allowed Texas to import electricity from other states during emergencies — were not adopted. The decision to stay independent of federal regulation was reaffirmed politically.

The regulatory structure remained largely market-based. Generators are still rewarded primarily for electricity they actually deliver, not for being reliable through extreme weather, though some adjustments have been made.

And the deepest structural choice — whether Texas would be a state that prizes cheap, abundant electricity above almost everything else, or one that prizes resilience and consumer protection — remained, fundamentally, the choice it has always been.

It is tempting, especially for people who do not live in Texas, to tell a simple story about the freeze: a state with too much faith in markets and too little willingness to be regulated finally paid the price. Texans got what they bargained for.

This story is partly true. It is also too simple, and it is not fair.

Texas's regulatory choices have produced real benefits for many decades. Cheap electricity. Easy entry for new generators. The fastest wind-power buildout in the country. Texas now produces more wind energy than any other state — more than several whole countries — partly because the same light regulatory hand that left the grid vulnerable in February 2021 also let wind farms get permitted

and connected in months rather than years. The same trade-off cuts in both directions.

Texas's voters and politicians have consistently chosen this trade-off. They have done so with information from earlier freezes and earlier warnings. They could have chosen differently. They have chosen, repeatedly, not to.

This is the deeper truth that Winter Storm Uri laid bare. The Texas freeze was not really a story about an engineering failure. It was a story about a political choice, made over many decades, that produced an engineering outcome.

This is true everywhere. California's blackouts during wildfire season are a story about the choice to live in fire-prone areas while running aging electrical infrastructure. Germany's energy crisis after the 2022 invasion of Ukraine was a story about the choice, made decades earlier, to depend on Russian gas while shutting down nuclear plants. France's relatively stable electricity prices through that same crisis were a story about the choice, also made decades earlier, to build a fleet of nuclear reactors.

There is no grid anywhere in the world whose performance is separable from political and economic choices. The reliability you experience is not just an engineering outcome. It is a political outcome that happens to be engineered.

Several months after the freeze, on a sweltering August evening, the same Texas grid was operating at near-record demand. Air conditioners were running across the state. ERCOT was managing the load minute by minute. The frequency stayed near 60 hertz. The lights stayed on.

Somewhere in a Houston suburb, a family was eating dinner together. The mother had lost a son in February. The grid that had failed her in winter was, that summer, working perfectly. The same engineers, the same control room, the same set of choices.

She set the air conditioning a degree cooler. She watched her remaining children at the table. The lights did not flicker.

The grid is a machine. The grid is also a mirror. It shows you, over time, the priorities of the people who built it.

In February 2021, the Texas grid showed something that people in Texas — and people watching from elsewhere — did not love seeing. But it was true. It is still true.

And the only way that picture changes is by changing the choices underneath it. Slowly. Imperfectly. By voters and engineers and legislators and ordinary people, deciding what they are willing to pay for what they want to get.



CHAPTER NINE

The Bare Roof



I'll be honest. The first time I really understood how solar panels worked, I started looking at every roof I drove past with a small, irritated voice in my head.

"Why isn't *that* one covered?"

It was hot. It was sunny. The sun was free. Solar panels keep getting cheaper. Anyone with a roof should be able to put a few panels up and shave a chunk off their electric bill, right?

For years, I assumed that any roof that did not have solar panels on it must belong to someone too lazy, too disorganized, or too uninterested to do the obvious thing.

I was wrong. The story is more interesting than that.

This chapter is about the bare roof. About all the reasons a roof you would expect to have solar might not. About why "just put up some panels" is one of those phrases that sounds simple until you actually try to do it.

Most of all, this chapter is about a deeper truth that comes up over and over in this book: **rooftop solar is not really a homeowner question. It is a grid question.**

Let's start with the dream. Then we'll work through what makes it harder.

The sun delivers, on average, about a kilowatt of power per square meter to the ground in a sunny place. A typical American house has 1,500 to 2,500 square feet of roof. Even if you only cover half of that with panels, even at modern panels' modest 22 percent efficiency, the math works out. A well-positioned American home could generate, over the course of a year, more electricity than the family inside it uses.

The hardware to do this has fallen in price by more than 90 percent in the last fifteen years. A 6-kilowatt system that cost \$50,000 in 2010 might cost \$15,000 today, before tax credits. That is a real, dramatic price drop. There has been no comparable price drop in any other major piece of household equipment, ever.

So the dream is: a sunny roof, modern cheap panels, free sunlight. The homeowner makes their own electricity, sells any extra back to the grid, and watches the bill shrink.

That dream is not impossible. Millions of American homes have it. In some neighborhoods of California, Arizona, Hawaii, Texas, and Florida, the majority of single-family houses now have solar. The dream works.

But the bare roof next door to a solar-covered one is not necessarily owned by someone foolish or lazy. It is more likely the product of a small stack of real constraints, any one of which can be the deal-breaker.

Let's walk through them.

Not all roofs are good roofs for solar.

Orientation matters a lot. A south-facing roof in the United States gets the most sunlight. East and west are okay. A north-facing roof is, for solar purposes, almost useless. If your house faces the

wrong direction, the big obvious-looking roof from the street is actually the back of the house, and the back of the house faces north.

Shade matters even more. Solar panels are surprisingly intolerant of shade. A single tree branch shading a single panel can drop the output of a whole string of panels significantly. Newer panels with built-in optimizers help, but they cost more. A neighbor's tall oak tree, planted thirty years ago by someone who did not anticipate solar panels, can make the whole project not worth doing.

Roof age matters too. Solar panels are heavy. They last 25 to 30 years. If your shingles have only 10 years of life left when the panels are installed, you are about to have a painful problem. You will either have to remove the panels, reroof, and reinstall them — adding thousands of dollars to the project — or live with leaks. Most installers will tell you that if your roof has less than 15 years left, you should reroof first and install solar second. Adding \$10,000 to a \$15,000 project changes the math considerably.

Roof material matters in a smaller way. Asphalt shingle roofs are easy. Tile roofs are harder. Metal roofs are easy. Wood shake roofs are a fire risk and may not be permitted with panels at all.

So just from the roof itself, before any of the other stuff, a meaningful fraction of American homes are simply not great candidates for solar. The neighbor with the bare roof might have a north-facing house, or a roof from 2003, or an oak tree.

Here is a constraint nobody thinks about until they have lived through it.

The person who owns the roof is often not the person who pays the electric bill.

About 36 percent of American households rent. Renters do not own their roofs. They cannot install solar panels even if they want to. Their landlord *could* install panels, but the landlord does not pay the electric bill, so the savings would go to the renter, not to the

landlord. The landlord pays the construction cost and gets nothing back. So the panels do not get installed.

This is called a **split incentive** in economics. The person who could pay for the upgrade does not benefit from it. The person who benefits cannot pay for it. The economically optimal outcome — install the panels, share the savings — requires a complicated arrangement that almost never gets made in practice.

The same split-incentive problem affects most commercial buildings (the building owner is not the tenant), most condominiums (the homeowners' association has to agree), and a lot of multi-family housing.

In some states and cities, programs exist to bridge this gap. **Community solar** lets renters and people with bad roofs buy a share of a solar farm somewhere else, and get the credit on their electric bill at home. It works. But it is much less common than rooftop solar and tends to be a smaller economic benefit.

The bare roof down the street might belong to a landlord who would be financially crazy to install panels they cannot profit from.

Now we get to the part that determines, more than anything else, whether a sunny roof in a sunny state has solar on it.

The rules that decide what happens to the electricity your panels generate.

A solar panel system on a house is connected to the grid. During the day, when the sun is up and the panels are producing, the system may generate more electricity than the house is using at that moment. The extra has to go somewhere. It flows back out through the meter, onto the wires in the street, and into the grid.

The big question: does the utility *pay you* for that extra electricity? And if so, how much?

For most of the last twenty years, the answer in the United States was: yes, the utility pays you, at roughly the same rate it would

charge you for buying it. This is called **net metering**. Your meter literally spins backward. If you generated more electricity than you used over a billing period, the utility owed you money.

Net metering was the policy that made rooftop solar economic for ordinary households. It is the reason California, Arizona, Massachusetts, and New Jersey became solar leaders.

But net metering has a problem. From the utility's point of view, it is paying retail prices for electricity it could buy on the wholesale market for a fraction of that. Worse, it is paying retail prices for electricity that arrives mostly at noon — when, as we learned in Chapter 6, there is already too much solar on the grid and prices on the wholesale market are sometimes negative. The utility is buying noon-sun at dinner-time prices, then trying to sell it to other customers at dinner time too. The math does not work for the utility.

So utilities have, for over a decade, been pushing to reduce or eliminate net metering. The fight has played out state by state, regulator by regulator.

In California, the fight came to a head in 2023. The California Public Utilities Commission replaced the previous net metering policy with a new one called **NEM 3.0** (Net Energy Metering, version 3). NEM 3.0 took effect on April 14, 2023.

Under NEM 3.0, a California homeowner who exports a kilowatt-hour of solar to the grid is no longer paid retail rate. They are paid a much smaller amount based on something called the **Avoided Cost Calculator** — essentially, the wholesale value of the electricity to the grid at that exact moment. For solar exported at noon, when the grid is flooded, this rate can be 70 to 80 percent lower than the retail rate.

A California homeowner who used to get about thirty-six cents per kilowatt-hour for exported solar now gets about eight cents.

The effect was immediate. Rooftop solar sales in California fell by between 66 and 83 percent in 2023 compared to 2022. By the end of

2023, the state had lost about 17,000 solar-industry jobs — 22 percent of California's solar workforce. Thirty-two solar companies in California went out of business in 2023 and 2024, including ADT Solar, one of the country's largest.

Three years later, in March 2026, after multiple lawsuits, the California Court of Appeals upheld NEM 3.0. The policy is, for now, the law.

Different states have different rules. Many states still have something close to full net metering. Some have moved toward versions of net billing similar to NEM 3.0. Arizona moved away from full net metering in 2017. New York and Massachusetts still have strong programs. Florida tried to weaken its program; the governor vetoed the change. The map keeps changing.

The honest summary is this: **two identical houses, on identical streets, with identical sunlight, can have wildly different solar economics depending on which side of a state border they're on.** And those state borders are themselves political, not natural. A roof in San Diego does not have solar for very different reasons than a roof in Phoenix.

There is one more constraint that almost nobody thinks about, and it deserves its own moment because of how surprising it is.

Sometimes the local distribution transformer — the gray canister on the pole or the green box in the yard, from Chapter 5 — cannot accept any more solar.

Here is why. A neighborhood transformer was designed to send electricity one way: down from the wires on the pole into the houses. When solar panels are exporting electricity *back up* through the transformer at noon, the transformer is being used in a way it was not originally designed for. There is a limit to how much backflow it can safely handle.

When that limit is reached, the utility will not approve more solar installations on that transformer's circuit. The houses on that circuit are simply told no. The utility has to upgrade the transformer first, which costs money and takes time, and which usually does not happen until enough houses on the block are demanding it.

This is happening more and more in older neighborhoods with high solar adoption. The bare roof next door to four houses with solar might be bare not because the homeowner does not want panels. It might be bare because the utility said no, and the homeowner does not know how to fight that fight.

Add to all of this the constraint we already met in Chapter 5: the service panel and wiring inside the house.

A solar installation requires that the house's electrical panel can accept the connection. Old houses with 60-amp or 100-amp service may need a panel upgrade before solar can even be installed. A panel upgrade can cost three to seven thousand dollars and take months to schedule because of the shortage of electricians in many parts of the country.

Add this to the cost of the solar system itself, and the payback math can shift from "ten years" to "twenty years." Most people will not invest in a twenty-year payback unless they plan to stay in the house for at least that long.

The bare roof might be bare because the panel inside the basement is original to a 1962 build.

So we have a stack.

The roof itself (orientation, age, shade). The owner-occupant mismatch (renters, condos, commercial). The state policy (net metering rules). The local transformer (saturated or not). The house's own wiring and panel.

Any one of these can be a dealbreaker. Most bare roofs are bare because at least one of them applies. A surprisingly large number of bare roofs have *all* of them applying at once.

This is what I mean when I say rooftop solar is a grid question, not just a homeowner question.

The choice to put solar on your roof is partly yours, yes. But it is also a choice by your state regulator about how exported solar is compensated. It is a choice by your utility about how much they have invested in the local transformer. It is a choice by your landlord, or your spouse, or your homeowners' association. It is constrained by the age of your roof and the angle of the sun and the trees your neighbors planted forty years ago.

When you see a bare roof, you are not necessarily looking at a failure of individual will. You are usually looking at a system that has not yet been built — or, in California's case, a system that was rolled back.

Rooftop solar is genuinely good. The sun is free. The panels work. Millions of American homes are now generating most of their own electricity. In sunny states with good policy, the payback is reasonable and the climate benefit is real.

But rooftop solar is not the *whole* answer to a clean grid. It is one piece of a much larger system that includes utility-scale solar farms in the desert, utility-scale wind farms on the plains, transmission lines linking them, storage to buffer them, and the grid management we have been learning about throughout this book.

In many places, utility-scale solar is now cheaper per kilowatt-hour than rooftop solar — meaning it is more efficient to build big solar farms in the desert and ship the electricity to your house than to install panels on your roof. The rooftop is more visible, more empowering, and more local. The desert farm is cheaper and more efficient. Both are true.

The bare roof, in other words, is not always a problem to be fixed. Sometimes it is the rational outcome of a system that has decided, correctly, that the electricity is better generated somewhere else.

The next time you drive through a sunny neighborhood, you can look at the bare roofs differently.

Some of them are bare because the homeowner has not gotten around to it. Some are bare because the homeowner is a landlord whose tenants pay the electric bill. Some are bare because the state changed the rules. Some are bare because the transformer on the pole next door is already full. Some are bare because the wiring inside is older than the moon landing.

And some of them are bare because somewhere out in the desert, a solar farm has already been built — to make electricity more cheaply than the bare roof ever could — and a transmission line is bringing it to that very house.

The roof is just one place where the choice gets made. The choice itself happens at a much bigger scale.



CHAPTER TEN

The Atom Comes Back



If you drive south on Pennsylvania Route 441 along the Susquehanna River, between Harrisburg and Lancaster, you will eventually pass four enormous concrete cooling towers rising out of an island in the middle of the river.

The towers are pale gray. They taper inward at the waist and flare out at the top like wineglasses the size of football stadiums. Two of them have been silent since 1979. The other two have been silent since 2019.

This is Three Mile Island.

For most Americans alive today, the name itself means an accident. On March 28, 1979, one of the reactors at this plant — Unit 2 — suffered a partial meltdown. A combination of a stuck valve, a faulty instrument reading, and operator confusion led to part of the reactor core melting inside its containment vessel. The accident did not kill anyone. The amount of radiation that escaped to the surrounding area was, according to most studies done since, not enough to cause measurable health effects. But the country's faith in nuclear power did not survive intact.

In the days after Three Mile Island, the American nuclear industry stopped. Power plants under construction were canceled. Plants on the drawing board were abandoned. Public opinion turned against new reactors. Anti-nuclear protests grew. For the next four decades, the United States built almost no new nuclear power.

The plant's other reactor, Unit 1, was unaffected by the 1979 accident. It kept running, safely, for another forty years. In 2019, its owners shut it down because it could not compete economically with cheap natural gas. The cooling towers went cold.

This chapter is about why those towers, in the spring of 2026, are getting ready to come back to life.

On September 20, 2024, the energy company Constellation announced a deal that would have seemed impossible a few years earlier.

Constellation owned Three Mile Island Unit 1 — the reactor that had been shut down five years before. They had signed a contract with Microsoft. A twenty-year contract. Under the contract, Constellation would spend roughly \$1.6 billion restoring Unit 1 to working condition. Microsoft would buy every kilowatt-hour of electricity the plant produced, for twenty years.

The reactor would be renamed the **Crane Clean Energy Center**, after a former Constellation CEO. It would have a planned restart in 2027 or 2028, pending regulatory approval. Its operating license would be extended through at least 2054.

The reason Microsoft wanted nuclear power, specifically, was its data centers. Microsoft is in an arms race with Google, Amazon, and a handful of other companies to build out the infrastructure that runs the world's artificial intelligence. Each new data center can use as much electricity as a small city. And Microsoft, like its competitors, has made public commitments to power its operations with

carbon-free electricity, twenty-four hours a day, every day of the year.

Solar and wind cannot do this on their own. They generate when the weather lets them, not when the data center needs them. Microsoft could buy enough renewable energy *on average* to cover its annual usage, but during the hours when the sun was not shining and the wind was not blowing, the data center would still be drawing power from the grid — which would still, in most places, be partly fossil-fueled.

Nuclear is different. A nuclear plant runs at full power for nine years out of ten. It produces almost no carbon emissions over its operating life. It is the only proven, scalable source of electricity that is both around-the-clock and low-carbon, on the timescale Microsoft and its competitors need.

So Microsoft did the math, and decided to pay for a closed nuclear plant to come back.

In November 2025, the Trump administration announced a one-billion-dollar federal loan to support the project. The plant is now expected to come back online sometime in 2027 or 2028.

Three Mile Island, the site whose name once stopped American nuclear power in its tracks, is one of the reasons American nuclear power is restarting.

The Crane Clean Energy Center is not the only deal of its kind. It is one of about a dozen significant nuclear announcements in the last three years, and it is part of a much larger shift.

Three pressures have been building up at the same time.

First, the demand for electricity is exploding. For nearly twenty years, US electricity demand was almost flat — efficiency gains canceling out growth. That changed around 2022. Data centers, electric cars, heat pumps, reshored chip and battery factories — all of them want more electricity, all at once. Total US demand is projected to

grow 25 percent by 2030. Some of the new demand is so concentrated geographically (Northern Virginia, central Ohio, eastern Texas) that local grids are already straining.

Second, climate goals require firm clean power. Cleaning up the grid using only solar, wind, and batteries is harder than it looks. The intermittency problem from Chapter 6 means you need enormous amounts of either storage or backup, both of which are expensive at scale. A growing number of engineers and policymakers have concluded that getting to a clean grid almost certainly requires *firm* clean sources too — sources that run regardless of weather. The two main candidates are nuclear and enhanced geothermal. Geothermal is promising but small. Nuclear is proven but politically and economically scarred.

Third, the economics have shifted. Restarting a closed nuclear plant turns out to be much cheaper than building a new one. The plant is already there. The transmission lines are already there. The skilled workforce can be rehired. The license is renewable. For about one to two billion dollars, you can bring back almost a gigawatt of zero-carbon, round-the-clock electricity in three or four years. Compared with the cost and timeline of building a new reactor — we will get to those numbers in a moment — restarts are a bargain.

These three pressures, together, are what have made 2024 and 2025 the most consequential years for American nuclear power since the 1970s.

Before going further, let's be straight about what nuclear is and is not.

Nuclear is one of the cleanest sources of electricity humans have invented, measured by carbon emissions over the entire life of the plant. Studies put nuclear's lifecycle emissions at around 12 grams of CO₂ per kilowatt-hour, roughly comparable to onshore wind, and far below natural gas (around 490) or coal (around 820).

Nuclear runs almost all the time. The average US nuclear plant operates at over 90 percent of its maximum capacity. It takes a few weeks off every year and a half for refueling, and that is about it. No other source of electricity comes close to this kind of reliability.

Nuclear is dense. A single one-gigawatt nuclear reactor occupies maybe a few hundred acres of land. To produce the same amount of electricity over a year with solar would require roughly 30,000 acres. To produce it with wind would require roughly 80,000 acres.

Modern reactor designs are dramatically safer than the ones built in the 1960s and 1970s. Passive safety systems use gravity and natural circulation to cool the core even if all the pumps and electricity fail. Newer fuel forms — particularly TRISO fuel, in which tiny pellets of uranium are coated in ceramic layers — can withstand much higher temperatures without releasing radiation.

Nuclear waste is a real problem, but a smaller one than it sounds. All of the spent fuel ever produced by the entire US nuclear fleet, over more than sixty years, would fill a single football field about ten yards deep. Compared to the waste streams from coal (millions of tons of ash, mercury, and CO₂ per year) or even from solar (lots of glass and silicon to recycle), nuclear waste is volumetrically tiny. It is, however, dangerously radioactive for tens of thousands of years, and the United States has not yet built a permanent repository to store it. The waste sits, at every operating nuclear plant, in concrete-and-steel casks on site.

Now the other side.

Nuclear plants are expensive to build, and the United States is bad at building them. The clearest case study is in Georgia.

The Vogtle Electric Generating Plant, near the town of Waynesboro, has had two operating nuclear reactors since the late 1980s. In 2009, its owners broke ground on two more: Units 3 and 4. They

were the first new commercial nuclear reactors started in the United States since 1979.

The project was originally projected to cost \$14 billion and to be completed in 2016 and 2017. It cost more than \$30 billion. It was completed seven years late. Unit 3 came online in July 2023. Unit 4 came online in April 2024.

The reactor designer, Westinghouse, went bankrupt during construction. The cost overruns will be paid for by Georgia Power customers, in the form of roughly \$14 to \$20 per month on their electric bills, for decades.

Defenders of nuclear say that Vogtle was a first-of-a-kind project, that the next plant of the same design would be cheaper, that we lost the muscle memory of building reactors during the four-decade pause. They are not entirely wrong. China and South Korea build similar reactors faster and at a fraction of the cost, suggesting it can be done. But the Vogtle experience is what it is. It is what every new American nuclear project has to argue against.

Nuclear waste storage has been politically stuck for forty years. The Yucca Mountain repository in Nevada was designated as the permanent national waste site in 1987. After billions of dollars of study and partial construction, it was suspended in 2010 over political opposition. As of 2026, the United States still has no permanent repository for high-level nuclear waste. The waste continues to accumulate at every operating plant.

Accidents have been rare but psychologically enormous. Three Mile Island in 1979 killed nobody but ended a generation of American nuclear construction. Chernobyl in 1986, in the Soviet Union, killed dozens directly and likely thousands more from cancer, contaminated thousands of square miles, and is still being managed today. Fukushima in 2011, after the Tōhoku earthquake and tsunami, displaced hundreds of thousands of people and led Germany

and Japan to make sweeping decisions about their own nuclear programs.

Modern reactors are far safer than the ones involved in these accidents. The newer designs being built now have passive safety systems that would have prevented all three. But the memory remains. Public support for nuclear in America has, however, shifted dramatically: a 2024 Pew Research poll found that 56 percent of Americans favored expanding nuclear power, up from 43 percent in 2020 — a 13-point shift in four years. Most Americans still favor solar and wind more strongly, but the gap is closing.

So that is the honest balance. Nuclear is clean, reliable, dense, and safer than it used to be. It is also expensive, politically fraught, and saddled with a waste problem we have not solved.

The most interesting bet in nuclear right now is that the future does not look like Vogtle.

For most of the history of nuclear power, the trend has been bigger. The reactors of the 1960s were a few hundred megawatts. The ones built in the 1970s and 1980s were a thousand megawatts or more. The two new ones at Vogtle are about 1,100 megawatts each. The logic was simple: economies of scale. Bigger plants are more efficient. Bigger plants spread the fixed costs over more output.

The trouble is that bigger plants also become enormous custom construction projects that take fifteen years to build and routinely run over budget. Each one is a first-of-a-kind. Each one requires assembling thousands of skilled workers in one place for a decade.

So a small group of nuclear engineers, starting around the late 2000s, had a different idea. What if the reactor were small enough to be built in a factory? What if instead of pouring concrete on a site for a decade, you trucked a reactor in?

These are called **small modular reactors**, or SMRs. They range from a few megawatts (small enough to power a remote town or an

industrial facility) up to about 300 megawatts (small enough that you can make several of them in parallel on an assembly line).

The bet is that what nuclear lost over the last fifty years was not the technology — the technology is fine — but the manufacturing experience. By making reactors smaller, simpler, and more standardized, you might be able to relearn how to build them, the way the aerospace industry learned to build jets.

Four American companies are leading the bet, each with a different approach.

NuScale, based in Oregon, has the most mature design. Each NuScale module is 77 megawatts and uses standard light-water reactor technology — the same kind that has powered submarines and nuclear plants for sixty years, just made smaller. NuScale is the only US SMR design that has received NRC design certification. In September 2025, the Tennessee Valley Authority signed an agreement to deploy up to **6 gigawatts** of NuScale capacity across its seven-state region — the largest SMR commitment in the country.

Kairos Power, based in California with reactor development in Tennessee, uses a more radical design. Its reactors are cooled by molten fluoride salt instead of water. The fuel is in the form of small ceramic-coated pebbles — TRISO fuel, the design that can withstand extreme temperatures without releasing radiation. Kairos has a partnership with Google to provide up to 500 megawatts of nuclear power by 2035. The first test reactor, called Hermes, is expected to operate at low power in 2026 in Oak Ridge, Tennessee.

TerraPower, founded by Bill Gates, is building a 345-megawatt sodium-cooled reactor called Natrium. The site is in Kemmerer, Wyoming — a former coal plant. Natrium pairs the reactor with a molten salt heat-storage system, so it can ramp output up and down to follow demand, which conventional reactors cannot easily do. The NRC is expected to issue a construction permit decision in the first

half of 2026. Meta has signed an agreement with TerraPower for up to eight Natrium plants. NVIDIA has invested.

X-energy, based in Maryland, is building a high-temperature, gas-cooled reactor called the Xe-100. Its first commercial deployment will be at a Dow Chemical facility in Texas. Amazon has invested in X-energy and signed for several reactors.

None of these reactors has yet produced commercial electricity. The earliest realistic dates for any of them are 2028 to 2030, with most deployments clustering in the early-to-mid 2030s. SMRs are a real bet, but they are still a bet. China and Russia, both ahead of the United States in deployment, have working SMRs. The United States does not, yet.

The most striking thing about the new American nuclear story is who is paying for it.

It is not, primarily, the federal government. It is not, primarily, the traditional utilities. It is — overwhelmingly — the technology companies.

In addition to Microsoft and the Crane Clean Energy Center, here are the other major deals as of 2026.

Amazon Web Services bought a data center campus directly next to the Susquehanna nuclear plant in Pennsylvania in March 2024 — the first deal in which a data center co-locates with a nuclear plant. Amazon also invested in X-energy.

Google signed a 500-megawatt deal with Kairos Power for advanced reactors by 2035.

Meta signed for up to eight Natrium reactors from TerraPower.

Oracle announced plans for data centers powered by three small modular reactors.

NVIDIA, the chipmaker whose GPUs power most of the AI industry, has invested in TerraPower.

OpenAI joined Amazon, Google, Meta, Microsoft, Oracle, and xAI in signing the Trump administration's "Ratepayer Protection Pledge" in March 2026 — committing to build, bring, or buy their own power and pay for the grid infrastructure upgrades their data centers require, rather than passing the cost to ordinary electric customers. The pledge is voluntary and not federally enforceable (electricity regulation lives mostly at the state level), but the public commitment is real.

By the spring of 2026, more than 10 gigawatts of nuclear capacity has been committed by hyperscalers and AI companies. Most of it will not come online for several more years. Some of it may never be built, if the technology fails to deliver or the projects fall apart. But the financial commitment is real, and it is the largest American investment in nuclear power since the 1970s.

Here is the part to sit with. For four decades, the case for new American nuclear power was made by environmentalists, scientists, and a small fraction of utilities, mostly without success. The case is now being made, and paid for, by the companies building artificial intelligence.

This is a genuinely strange alliance. It is also one of the most consequential energy stories of the decade.

It would be tempting to call this a "nuclear renaissance" and move on. The phrase has been used so many times over the last twenty years that it has lost most of its meaning.

The honest version is more careful.

American nuclear power is reopening some closed plants. It is building one type of large reactor — the AP1000 design at Vogtle — very slowly and very expensively. It is making serious progress on at least four different small modular reactor designs, none of which has yet produced commercial electricity. It is signing big-name deals that may or may not be built. And it is doing all of this against the

backdrop of a still-unsolved waste problem, a public memory of three major accidents, and a track record on cost that does not, in the American context, inspire confidence.

It is also doing it at a moment when the world genuinely needs more firm, clean electricity than anyone knows how to build. The demand from data centers alone, by 2030, will be larger than the entire current output of every nuclear plant in the United States combined. Renewable energy is being built faster than anyone predicted, but renewable energy alone cannot solve the problem if the goal is also reliable, around-the-clock, low-carbon electricity. Some new firm clean power is going to be built. Some of it will probably be nuclear. How much is the open question.

If you are reading this in middle school or high school, you will live through the answer. You will see the SMRs come online or fail to come online. You will see the Vogtle experience repeated or improved. You will see whether the Microsoft-Constellation kind of deal becomes standard or stays unusual. You will see, in your lifetime, whether the United States can build large clean infrastructure or whether the Vogtle pattern repeats.

A few years from now, if all goes as planned, the cooling towers at Three Mile Island will begin to vent steam again.

Most days, the steam will be a thin white plume rising off the towers' tops, dissipating into the Pennsylvania sky. The reactor inside Unit 1 will be running quietly, producing about 835 megawatts of electricity — enough to power something like 800,000 American homes, or, in the new arrangement, a chunk of the AI infrastructure that will help train and run the artificial intelligence used by hundreds of millions of people.

The plant will produce essentially no carbon emissions in operation. It will run 24 hours a day, 7 days a week, 365 days a year, with brief outages every eighteen months for refueling.

It will also be sitting next to Unit 2, the reactor that suffered a partial meltdown in 1979 and has been silent ever since. The two reactors will be on the same island in the Susquehanna River. The same towers will be visible from the same highway. The plant will have, in some ways, gone full circle.

Whether you think this is the right way to power the future, or the wrong one, or one of several reasonable ways, is a question this book is not going to answer for you. There are intelligent, honest people who believe nuclear power is the only realistic path to a clean grid. There are intelligent, honest people who believe it is dangerous and uneconomic and we should not build any more of it. The arguments are real on both sides. The trade-offs are real. The choices are being made right now.

The cooling towers, in any case, will be back. Whether they will be the start of a real American nuclear comeback, or a one-time symbol of a moment when tech companies briefly needed something only nuclear could deliver, will be one of the most important stories of the 2030s.

It is not yet decided. The thing about being twelve or fourteen years old when you read this book is that you may be the one who decides.



CHAPTER ELEVEN

The Grid That Has to Grow



In Loudoun County, Virginia, about forty miles west of Washington D.C., there is a stretch of suburban road that looks, at first glance, like an ordinary American office park. Big rectangular buildings sit behind landscaped berms. Parking lots are mostly empty. There are no signs identifying what is inside.

This is one of the most important places in the modern world.

The buildings are data centers. Inside them are millions of servers — the physical hardware that runs a substantial portion of the global internet. When you stream a movie, send an email, search for something, or ask an artificial intelligence a question, the request very likely passes through Loudoun County. The local industry trade group estimates that about 70 percent of the world's internet traffic touches a data center somewhere in northern Virginia. By most measures, Loudoun County hosts more data center capacity than any single place on Earth.

In 2025, Loudoun County's data centers drew about 4 gigawatts of continuous electricity — roughly the same amount of power as four large nuclear plants running flat out. The county's total demand for data center power has grown by about 30 percent per year for the

last several years. By 2030, the demand could be 12 gigawatts or higher.

To put this in perspective: the entire state of Virginia, a state of 8.6 million people, currently uses about 17 gigawatts at peak demand. The data centers in one county will, within a few years, demand more electricity than the rest of the state's homes, businesses, and industries combined.

Somewhere in an electric utility's office in northern Virginia, there is a planner trying to figure out where this electricity will come from. That planner has spent most of their career planning for an electric grid that did not need to grow.

Until very recently, no American electric planner had needed to plan for growth at all.

This may be the most surprising fact in this entire book.

From the year 2000 to the year 2020, total United States electricity demand was almost flat. It went up a little. It went down a little. Over the course of two decades, it grew by less than 5 percent in total.

Two things happened simultaneously to produce this strange equilibrium. The American economy kept growing. New houses kept getting built. New devices kept getting plugged in. Computing power exploded. By all the old logic of electricity demand, the country should have been using a lot more power year after year.

But it was not, because efficiency was getting better just as fast.

The shift from incandescent light bulbs to compact fluorescents and then to LEDs reduced the energy used for lighting by something like 80 percent over those twenty years. Refrigerators, washing machines, air conditioners, and televisions all became dramatically more efficient. Heating and cooling systems improved. Factories deployed better motors and better insulation. The country was do-

ing more with the same kilowatt-hours, year after year, and the net result was that the grid did not need to grow.

This was, generally speaking, considered a triumph. It meant that economic growth no longer required ever-increasing energy use. It meant that climate concerns could be addressed without slowing the economy. It also meant that the electric utility industry, for two decades, mostly stopped building new power plants. The plants from the 1970s and 1980s were still there. They were enough.

For an entire generation of grid planners, this was the world. No new generation needed. Transmission upgrades only where things were aging out. Utility executives became experts at managing flat demand, optimizing existing plants, and squeezing efficiency. They did not become experts at building.

Then, sometime around 2022, the world changed.

The flat-demand era ended fast.

By 2023, every credible forecast was showing demand growing. By 2024, the forecasts were getting steeper every quarter. By 2025, electric utilities across the country were filing emergency requests with state regulators to build new generation, to upgrade transmission, to raise rates to fund the build-out.

A leading consulting firm, ICF, calculated in 2025 that US electricity demand would grow **25 percent by 2030**, compared to 2023. Other forecasts put the growth higher. The International Energy Agency officially declared the world to have entered an "Age of Electricity," with global demand growing faster than at any time since the postwar build-out of the 1950s and 1960s.

Three forces are driving the growth.

The biggest single driver, by most estimates, is data centers — and particularly the data centers built for artificial intelligence.

In 2024, US data centers consumed about 183 terawatt-hours of electricity — about 4 percent of total US electricity consumption. By

2030, depending on whose forecast you read, that number could rise to 9 percent, or 11 percent, or even higher.

The reason AI specifically demands so much electricity is that training and running the largest AI models requires enormous amounts of computation. A single AI training run for a state-of-the-art model can consume as much electricity as several thousand American homes use in a year. AI inference — actually running a trained model to answer your questions — uses less per query, but is happening billions of times per day across the world.

Globally, the International Energy Agency forecasts that data center electricity demand will grow from 415 terawatt-hours in 2024 to 945 terawatt-hours by 2030 — more than doubling in six years. The United States and China together will account for almost 80 percent of that growth.

Here is what makes this hard for the grid. Data centers are not spread out evenly. They cluster in specific places — Northern Virginia, central Ohio, eastern Texas, Phoenix, Arizona, and a handful of others. The cluster effect compounds the engineering challenge. A single new gigawatt-scale data center being built near an existing cluster does not just add a gigawatt to national demand. It adds a gigawatt to a *local* grid that may not have a gigawatt of spare capacity within reasonable transmission distance.

In Loudoun County alone, S&P Global's 451 Research projects that data centers will draw 12 gigawatts by 2030 — roughly tripling from 2024 levels. In Texas, demand is growing nearly as fast. The local utilities have to either generate this electricity nearby or build the transmission lines to bring it in from elsewhere. Often, neither option is fast enough.

The second force is everything else getting plugged in.

For the last decade, federal and state policy has been pushing American households and businesses to switch from fossil fuels to

electricity for as many uses as possible. Heat pumps to replace gas furnaces. Electric induction stoves to replace gas burners. Heat pump water heaters to replace gas tanks. Electric vehicles to replace internal-combustion cars.

Each of these changes saves carbon emissions, because most modern electricity is significantly cleaner than burning gas in your basement, and because the technology itself is more efficient. A heat pump can move three or four units of heat for every unit of electricity it uses. An electric car converts electricity to motion at over 90 percent efficiency, compared to a gasoline car's roughly 30 percent.

But the same shift also adds electric load. American homes that fully electrify use significantly more electricity than they did before — they have just stopped burning gas separately.

In the early 2020s, the rate of electrification accelerated. EV sales in the United States grew 10 percent year-over-year in early 2025. Heat pump sales grew 14 percent in 2024 and 9.5 percent in early 2025. Each of these adoptions, multiplied across millions of households, adds a steady increase to grid demand.

For many decades, grid planners in northern Virginia and Texas could ignore residential electrification because it was small and slow. By 2024, they could no longer ignore it.

The third force is the most recently visible: American manufacturing is rebuilding.

For decades, the United States offshored most of its industrial production. The trend reversed in the late 2010s and accelerated in the 2020s. The reasons are several — pandemic supply-chain shocks, US-China political tensions, government subsidies under the CHIPS Act and the Inflation Reduction Act, and the long-term recognition that critical industries should not all be located in any single foreign country.

The result has been a manufacturing construction boom, particularly for two categories: semiconductor fabrication plants (the "fabs" that make computer chips) and battery factories.

Semiconductor fabs are some of the most electricity-intensive buildings ever built. A single new TSMC fab in Arizona, or Intel fab in Ohio, can draw a gigawatt of continuous power — comparable to a small city. Battery factories are not far behind. Both are also highly concentrated geographically, like data centers, because they need huge supplies of trained labor and water and zoning permits, all of which are easier to assemble in some places than others.

By the first quarter of 2025, US construction spending on manufacturing facilities was more than double what it had been three years earlier. Most of those facilities, when they come online, will be heavy electricity consumers for the next several decades.

So that is the demand picture. Twenty-five percent growth by 2030, conservatively. Concentrated in a few places. Driven by data centers, electrification, and reshored manufacturing.

Where will the new electricity come from?

Every source we have talked about in this book is being deployed, fast.

Natural gas is, in the short term, the biggest beneficiary. Gas plants can be built faster than almost anything else — eighteen to thirty months from groundbreaking to operating — and they provide firm, dispatchable power. Several states are seeing new gas plants approved for the first time in years. This is controversial. Gas is a fossil fuel. Building new gas plants in 2026 means committing to those emissions for the next thirty or forty years. Some clean-energy advocates argue this is a betrayal of climate goals. Some utilities argue there is no other option that can be built fast enough. Both sides have a point.

Solar and wind continue to be built at record pace. The United States added about 50 gigawatts of new solar and wind capacity in 2024. These remain the cheapest sources of new electricity per kilowatt-hour, even after accounting for intermittency. But, as Chapter 6 showed, you cannot run a grid on solar and wind alone without storage and transmission — both of which are also being built, and both of which are still bottlenecks.

Nuclear is back, as Chapter 10 showed. The Three Mile Island restart, the Vogtle reactors, the SMR deals. But most of this capacity will not come online until the late 2020s or the early 2030s. In the immediate term, nuclear is a hope, not a solution.

Geothermal is the dark horse. New techniques — drilling deeper, using methods borrowed from the fracking industry to engineer underground reservoirs — are making geothermal viable in many more places than the few traditional hot-spring regions. It is firm, like nuclear, but cleaner-image and faster to build. Several large geothermal projects are now under construction in Nevada, Utah, and Texas.

Behind-the-meter generation — power plants built right next to the data center or factory they serve, bypassing the grid — is growing fast. Most of these are gas turbines, but some are nuclear (the Amazon-Talen deal from Chapter 10) and some, increasingly, are solar plus storage.

Across all of these together, the United States is building generation faster than it has built it in decades. But not faster than the demand is growing in some places. The gap is the political problem.

Even where new electricity is being generated, getting it to where it needs to go is increasingly the harder problem.

The International Energy Agency reports that more than **2,500 gigawatts** of generation projects are stuck globally in **grid connection queues** — that is, projects that are built or nearly built but cannot

connect to the grid because the transmission system cannot accept them. In the United States, the queue is so backlogged that some renewable energy projects are waiting four to seven years just to be studied for interconnection.

This is the long road problem from Chapter 3, made urgent. We have the engineering. We have the money. We have the demand. We do not have the wires. And we do not have a fast way to build the wires.

Some of the most ambitious responses to the demand surge are trying to *avoid* the transmission problem entirely. Hyperscalers are co-locating their data centers next to power plants. Manufacturers are building their own gas turbines on site. New startups are putting AI compute nodes on individual homes, where the local distribution grid can already serve them. We will talk about that in the next chapter.

There is one question this chapter has been circling that we have to address directly.

When a utility builds a new power plant or a new transmission line, the cost is recovered from electricity customers — in the form of higher rates on their bills. Traditionally, the cost has been spread across all customers in the utility's service territory. The logic was that everyone benefits from a reliable grid, so everyone should share the cost.

But when most of the new demand is coming from a handful of giant new data centers owned by some of the wealthiest companies in the world, that traditional logic starts to feel unfair. Why should a retiree on a fixed income in rural Virginia subsidize the transmission upgrade required to serve a Microsoft data center?

This question has become politically explosive in the last two years. Virginia, Texas, Ohio, and several other states have seen residents organizing against new data centers and against rate in-

creases. Polls show electricity affordability rising rapidly as a political issue. Energy prices nationally rose 6.3 percent in the year ending early 2026, according to the federal Consumer Price Index — faster than overall inflation.

In February 2026, President Trump used his State of the Union address to announce something he called the **Ratepayer Protection Pledge**. A few weeks later, in March 2026, seven of the largest American technology companies — Amazon, Google, Meta, Microsoft, OpenAI, Oracle, and xAI — gathered at the White House to sign it.

Under the pledge, these companies committed to "build, bring, or buy" all the new electricity their data centers will require. They committed to pay for any new infrastructure — transmission, substations, even local distribution upgrades — needed to serve them. They committed to negotiate separate rate structures with utilities so that the cost would be assigned to the data centers, not spread across all customers. They even committed to pay for the electricity whether they used it or not.

The pledge is voluntary. There is no federal enforcement mechanism. American electricity is mostly regulated at the state level, and states have varying rules about how costs can be allocated. Some energy-policy experts have called the pledge largely symbolic — a public commitment by tech companies to do things they were probably going to do anyway, dressed up as a political win.

But the pledge does reflect a real shift. The biggest tech companies do, in fact, want to be seen as good neighbors to the communities they operate in. They have learned that public backlash against data centers can slow their permits, raise their costs, and politicize their existence. The math, for the companies, is to pay for their own infrastructure even when the law does not strictly require it.

Whether the pledge will actually keep residential rates from rising is a question that will be answered over the next several years.

Electricity rate decisions are slow. Effects on bills, even if the companies fully follow through, will take at least two years to show up. The fight will continue in state regulatory hearings, in court cases, and in elections, for the rest of the decade.

The honest framing of all this is not "data centers are eating residential electricity." The grid is not a fixed pie. The data centers are paying for new generation that, in many cases, adds to total supply. Some of that new generation, particularly the renewable power and the nuclear restarts from Chapter 10, would not have happened without their demand.

The honest framing is this: **the grid is being asked to grow faster than it has in fifty years. Who pays for the growth, and how the costs are allocated, is a political question that will be litigated state by state, year by year, for as long as the demand keeps rising.**

Somewhere in northern Virginia, the planner is still at her desk, looking at the same forecast.

The forecast says her utility's service area will need to add somewhere between five and ten gigawatts of new electricity capacity over the next five years. That is, in five years, what utilities used to build in fifty.

She will not do it alone. The same forecast is being looked at by planners in Ohio and Texas and Arizona. The same calculation is being run by every electric utility in the country. The same supply chains for transformers, for transmission cable, for skilled electricians, are being stretched to their limits.

For two decades, her job was about efficiency. About squeezing more out of what was there. About making the grid leaner.

For the rest of her career, her job will be about growth. About building things the country has not built in a generation. About arguing with regulators, with neighbors, with environmental re-

viewers, with state legislators, about where the lines go and who pays for them.

This is the central engineering and political project of the next decade. It is not a problem that data centers caused, or that electrification caused, or that any single industry caused. It is a problem that the entire modern economy has, by deciding it wanted more electrification of everything, and more AI, and more reshored manufacturing, and a cleaner grid, all at once.

We chose all of these. We are going to have to build the grid that can serve them.

The planner closes her laptop and goes home. Tomorrow, the forecasts will be a little higher than they were today.



CHAPTER TWELVE

The Grid Grows a Brain



On a hot July afternoon in California, the grid is approaching its limits.

It is 4 PM. The temperature is in the high 90s across most of the state. Air conditioners are running flat out. Solar panels, after a long bright day, are beginning to lose their output as the sun drops lower in the western sky. The duck curve from Chapter 6 is rising into its evening neck. Power plants across the state are ramping up. Reserves are getting thin. The grid operators at CAISO are watching their screens carefully.

Twenty years ago, this would have been the moment they made an urgent phone call. *Bring everything you have online.* Reserve gas plants would have started spinning. If those were not enough, the operators would have started preparing rolling blackouts.

Today, something different happens. Across California, in homes the operators will never visit, signals begin to fan out through the internet.

A few thousand Nest thermostats receive a message to nudge their air-conditioner set point up by two degrees. Most homeowners do not notice. The houses stay comfortable.

About twenty thousand Tesla Powerwalls, sitting on suburban garages, begin discharging the solar electricity they have been storing all day. Each one is pushing about three or four kilowatts back onto the grid. The combined output, just from those Powerwalls, is roughly equivalent to a small natural gas plant.

Several hundred large industrial customers — manufacturing plants, cold storage warehouses, water treatment facilities — automatically reduce their non-critical loads in response to a pricing signal that just hit their energy management systems.

A few thousand electric cars, plugged in at homes and offices across the state, briefly stop charging. They will resume in an hour or two when the evening peak has passed.

The grid balances. The lights stay on. No emergency. No blackout. No one in the affected homes had to do anything except, possibly, glance at the thermostat and wonder why it suddenly said 76 instead of 74.

This is the smart grid. It is real. It is partially deployed. It is one of the most important shifts in how electricity works since the invention of the grid itself.

To understand why the smart grid matters, you have to understand how dumb the old grid was.

For most of the twentieth century, the electrical grid was a one-way pipe.

Power flowed in one direction: from generators (coal, gas, nuclear, hydro) through transmission lines, through substations, through distribution wires, into homes and businesses. The homes and businesses were passive consumers. They drew power when they wanted it. The grid had no way to know who was drawing what, except by reading meters once a month with a person walking around in a uniform.

The grid had almost no awareness of itself. The operators in the control room from Chapter 2 could see what their own generators were doing, and roughly what was happening at the major substations. They could not see, in real time, what was happening at the level of neighborhoods or individual customers.

When the lights went out, the utility usually found out because someone called.

This was not because the engineers were lazy. It was because the technology to do anything else did not exist for most of the grid's history. You cannot have a smart grid without computers, sensors, GPS time-synchronization, fast wireless communication, and electricity-cheap microchips. None of those things existed in 1930. Only some of them existed in 1990. All of them exist now.

The smart grid is what happens when you put modern computing, communications, and sensing onto a system that was originally built without any of those things.

The first major piece of the smart grid was the **phasor measurement unit**, or PMU.

A PMU is a sensor about the size of a shoebox. It sits inside a substation or at a key point on the transmission grid. It measures the voltage and current of the grid up to 120 times per second. And — this is the key part — every measurement is tagged with a GPS time stamp accurate to the microsecond.

GPS time stamps are what make PMUs powerful. With them, a PMU in California and a PMU in New York can compare their measurements down to the millisecond. Operators can see, in real time, exactly how the grid is behaving across the whole country. They can detect wobbles in frequency, voltage sags, oscillations that did not exist before — all of which used to happen invisibly, in the seconds before a cascade, with no one watching.

The push to install PMUs began seriously after the August 2003 blackout (Chapter 4). One of the lessons of that disaster was that the grid had been blind. The operators in Ohio could not see what was happening in their own system, let alone across state lines. A US Department of Energy program began funding PMU deployment in the late 2000s. By 2026, a typical American electric utility has dozens to hundreds of PMUs streaming data into its control room, every minute of every day.

If you remember Chapter 2's comparison of the grid to a heartbeat, PMUs are what allow operators to see the heartbeat clearly for the first time. Where the older system gave them x-rays — useful but slow — the PMUs give them an electrocardiogram, updating in real time.

The second major piece of the smart grid is the **smart meter**.

You probably have one. About three-quarters of American households do, as of 2026.

A smart meter is the small device, usually mounted on the outside wall of your house near where the electric service comes in, that has replaced the old analog spinning-wheel meter. It measures your electricity consumption every 15 minutes to one hour and reports the readings back to the utility automatically over wireless or power-line communications.

Smart meters do three important things.

First, they let the utility see, in real time, exactly which houses are using how much electricity. When a tree branch falls on a line and your block goes dark, the utility now knows immediately, without anyone calling. This shortens outage response times dramatically.

Second, they enable **time-of-use pricing**. Instead of paying a flat rate per kilowatt-hour all day, your bill can vary based on when you used the electricity — cheap at midnight, expensive at 6 PM. This gives the utility a tool to shift demand off the peak hours when the

grid is most stressed, and it gives you a tool to save money by doing your laundry overnight or programming your dishwasher to start at 2 AM.

Third, they let the grid talk to your house.

That last one is the most important, and we will spend the rest of the chapter on it.

Here is the most important shift in modern grid management.

For a hundred years, the grid had one tool: it could ramp generation up and down to match demand. Need more power at 6 PM? Run more gas turbines. Less power at 3 AM? Take some plants offline. The grid worked supply.

The smart grid added a second tool: it could also flex *demand* to match generation. Need less power at 6 PM? Pay customers to use less. Got too much solar at noon? Pay customers to use more. Suddenly the grid had two ways to balance, not one.

This is called **demand response**, and it is one of the most powerful and least understood tools in modern grid management.

Demand response works because most electricity use is somewhat flexible if you give people the right signal. Air conditioning does not have to run at exactly 100 percent at 5:30 PM — the house can coast for fifteen minutes at a slightly higher temperature without anyone noticing. Water heaters do not have to run on a fixed schedule — they can heat up at midnight when the grid is sleepy and hold the heat. Electric cars do not have to charge the moment you plug in — they can wait until the cheapest hour of the night.

In practice, demand response programs work in two main ways.

In a **price-based** program, your utility charges you different rates at different times. You see the rates, you adjust your behavior, the grid benefits indirectly. This is the simplest version.

In a **direct-control** program, you give the utility permission to nudge your devices on your behalf. A smart thermostat. A smart

wa-

ter heater. A smart EV charger. The utility sends a signal that says *the grid is stressed for the next hour — please cycle off*. Your house cooperates automatically. You may not even notice. At the end of the month, you get a small credit on your bill for participating.

The number of demand-response programs in the United States has more than doubled in the last five years. Most large utilities now have at least one. The collective load they can shift — what engineers call "controllable load" — is growing into the tens of gigawatts nationally. That is, the entire grid's daily peak can be moved by tens of gigawatts, just by sending signals to enough smart devices.

This is real, and it is happening now. Your refrigerator is starting to talk to the grid. The grid is starting to talk back.

Once homes have smart meters and demand-response devices, the next shift becomes possible: homes that not only consume electricity but also *produce* it.

A house with rooftop solar generates electricity during the day. Some of it the family uses; some of it flows back to the grid. A house with a battery (Chapter 7) can store excess solar from noon and discharge it during the evening peak. An electric car parked in the driveway can, with the right hardware, charge from the grid when prices are low and feed back to the grid when prices are high.

These small generators and storage devices, scattered across millions of homes, are called **distributed energy resources**, or DERs.

In the old grid, the distribution network — the wires from the substation to your house — was a one-way pipe. Power flowed out to customers. Period.

In the new grid, the distribution network is becoming a marketplace. Power flows in both directions. The local transformer (Chapter 5) sometimes pushes power down to houses, sometimes accepts

power back up from houses, often in the same hour. The wires were not originally designed for this. The local utilities are upgrading them — sometimes faster than they would like, sometimes slower than the homeowners would like — to handle the new bi-directional flow.

When you coordinate enough DERs across enough homes, something remarkable becomes possible.

If you have ten thousand Tesla Powerwalls scattered across ten thousand California homes, you have a strange asset. Individually, each Powerwall stores 13.5 kilowatt-hours of electricity. Collectively, they store 135 megawatt-hours. If you could discharge them all at once, you could deliver 50 or 60 megawatts of power for two or three hours — equivalent to a medium-sized power plant.

The catch is that until recently, you could not coordinate them. Each Powerwall belonged to a different homeowner, on a different street, with different settings.

Now you can. Companies like Tesla, Sunrun, and Sunnova have built software systems that aggregate thousands of distributed devices and bid their combined capacity into wholesale electricity markets — exactly as if they were a single power plant. These aggregations are called **virtual power plants**, or VPPs.

When the grid is stressed, the VPP operator sends a signal. Thousands of home batteries discharge in coordinated unison. Thousands of smart thermostats nudge their settings. Thousands of EV chargers pause. The combined effect is the equivalent of bringing a 50- or 100-megawatt power plant online — but the plant does not exist as a physical building. It exists as a contract between a software company and ten thousand homeowners.

Virtual power plants are real and growing. By 2026, the United States has several gigawatts of VPP capacity operating, with much

more planned. California, Texas, and Vermont have been particularly aggressive in encouraging them.

This is the deepest change in the grid since alternating current. The biggest power plants in the country, ten years from now, may not be physical buildings at all. They may be aggregations of small distributed resources, coordinated in software, scattered across millions of homes.

There is one more layer worth understanding.

A traditional distribution network responds to a fault — say, a tree branch falling on a line — like this: the protective equipment at the substation detects the fault. The circuit breaker opens. A neighborhood goes dark. A utility crew is dispatched. Hours later, the crew finds the fallen branch, removes it, closes the breaker, and the lights come back.

A smart distribution network responds differently. The same tree branch falls. The same fault is detected — but now, automated switches scattered along the distribution circuit decide, in seconds, to isolate the small section where the fault occurred and reroute power around it. Most customers see a flicker, not an outage. The crew is still dispatched to fix the branch, but the lights stay on while they work.

This is called **distribution automation**. The most-cited American example is in Chattanooga, Tennessee. The local utility, EPB, invested heavily in smart switches and fiber-optic monitoring across its entire system starting in the late 2000s. After the investment, the average outage time in Chattanooga dropped by roughly 50 percent. During a major windstorm, the system rerouted around dozens of faults automatically, keeping most customers from ever seeing an outage.

Most American utilities are still in the early stages of distribution automation. The hardware works. The economics depend on how

the costs of the upgrades are recovered, which is a state-by-state regulatory question. As with much of the smart grid, the bottleneck is no longer the technology. It is investment and regulation.

Now we come to the most extreme version of all of this — and it is happening right now.

On April 14, 2026, a San Francisco company called Span announced a partnership with Nvidia and the homebuilder PulteGroup. The product they unveiled, called **XFRA**, takes the trend of distributed grid resources to a logical extreme.

An XFRA unit is a small box, about the size of a large kitchen appliance, that gets mounted on the outside wall of a newly built home. Inside the box is a Dell server containing sixteen Nvidia RTX Pro 6000 Blackwell GPUs and four AMD EPYC CPUs — about three terabytes of RAM and roughly the same compute as a small data center rack. The whole thing is liquid-cooled. It draws between one and two kilowatts of continuous power, depending on workload.

The XFRA unit connects to the home through a Span smart panel — the same kind of smart panel that some homeowners install to manage their solar and EV chargers. The smart panel measures the home's electricity usage minute by minute and determines how much spare capacity is available on the home's electrical service. The XFRA unit uses only that spare capacity. When the home needs more power — when the AC kicks on, or someone starts charging an EV — the smart panel throttles the XFRA unit down.

The XFRA unit runs AI workloads. Not training the giant models — the latency from a residential internet connection is too high for that. Instead, it runs *inference* — the actual day-to-day work of answering people's AI queries. A user somewhere on the internet types a question to an AI service. The service routes the question to the nearest available compute node. That node, increasingly, might be a box on the outside wall of a house in Phoenix or Dallas.

The homeowner gets a flat fee for hosting the unit. Their electricity bill is reduced or covered. Their internet is paid for. The Span panel ensures the XFRA never interferes with the home's own electricity needs.

In May 2026, the system was deployed in exactly one home — a Pulte demonstration installation. If the pilot works, Span says they could deploy thousands of units in PulteGroup's roughly one thousand active home communities across the United States. Their pitch is that they can roll out the equivalent of a 100-megawatt data center in distributed form at roughly one-fifth the cost and six times the speed of building a centralized facility.

Whether XFRA works at scale is a real open question. The reliability of residential internet, the willingness of homeowners to host the boxes, the cybersecurity risks of putting compute hardware on the outside of strangers' houses — none of these are settled. The pilot is real but small.

What is striking, regardless of whether XFRA succeeds, is the direction it points. For a century, the home was a passive consumer of electricity. In the last fifteen years, some homes have become small generators (rooftop solar) and small batteries (Powerwalls). Now some homes are becoming small data centers, too.

The same wire that brings you electricity, increasingly, can also carry away your spare electricity, your spare bandwidth, and your spare computing power, and turn them into income.

The house, in the smart grid era, is no longer a destination. It is becoming a node.

The smart grid is real. The smart grid is partially deployed. The smart grid is one of the most consequential technological shifts of the 21st century, even though most of it is invisible to the people benefiting from it.

But it is worth saying clearly: this is not a finished story.

Most American utilities have deployed only some of these technologies, and unevenly. Smart meters are widespread but not universal. Demand response programs exist but are nowhere near their potential scale. Virtual power plants are growing but small relative to the traditional grid. Distribution automation is the exception, not the rule. XFRA is in one house.

There are also genuine concerns worth naming.

Privacy. Smart meters reveal a lot about you. The pattern of your electricity usage tells someone with access to the data when you are home, when you go to bed, what appliances you own, what your daily routine is. Most utility data is protected, but breaches happen, and the data can be subpoenaed in legal cases.

Cybersecurity. A digital grid has a much larger attack surface than a non-digital one. A foreign government, or a criminal organization, that takes over a virtual power plant could cause real damage. The Russian attacks on the Ukrainian grid in 2015 and 2016 showed that this is not theoretical. The American grid has so far avoided major cyber incidents — but the systems are getting more complex faster than the defenses are getting more robust.

Equity. Smart-grid benefits flow disproportionately to households that can afford smart thermostats, EVs, solar panels, and home batteries. Renters, lower-income households, and people in older buildings often cannot participate fully. The risk is that the smart grid becomes a luxury feature of wealthy neighborhoods, while older neighborhoods get the same old service.

These are real concerns, and the people building the smart grid are taking them seriously — but not always seriously enough. They will be the subject of policy fights for the rest of this decade.

Throughout this book we have described the grid as a giant synchronized machine, a juggling act, an invisible network connecting everything. All of those framings are still true.

The smart grid adds one more.

The grid, increasingly, has a brain.

It senses itself in real time, all the way down to individual neighborhoods. It talks to thousands of small assets in millions of homes. It coordinates them automatically, without anyone explicitly telling it to. It heals around its own faults. It is starting to host new kinds of activity — compute, AI, distributed services — on its physical infrastructure.

The grid of 1980 was a machine. The grid of 2030 is becoming something closer to a nervous system.

If you have ever wondered why your Nest thermostat sometimes adjusts itself before you do, or why your utility offered you a credit to enroll your water heater in a program, or why a new house in Arizona has a strange-looking box on the outside wall — you are seeing pieces of this. The pieces are still being assembled. The picture is still emerging. But the shape of it is clear enough now.

The grid is not just a way of delivering electricity anymore. It is a network. And like all networks, the most interesting thing about it is what people will choose to put on top.



CHAPTER THIRTEEN

The Order of Operations



The single most common question a homeowner asks when they start thinking about clean energy is: "Should I get solar?"

This is the wrong question.

Not because solar is bad. Solar is genuinely good. Millions of American homes have solar panels and have reduced their electric bills and their carbon footprint as a result. But solar is, almost always, the *last* major step a household should consider in clean-energy upgrades, not the first.

The reason this matters is that most homeowners only have the time, money, and attention to make a few big changes. If they spend their budget on the visible thing — the panels on the roof — they often run out of energy and money before they get to the less visible things that would have done more for them, and for the planet.

This chapter is about the right order of operations. It is about what an honest, thoughtful homeowner — or someone advising one — should think about first, second, third, and last.

It is also, indirectly, about a deeper lesson that comes up over and over in this book: **the most important things are often invisible.**

Here is the honest order, cheapest carbon reduction per dollar first.

- Seal the leaks. Insulate.
- Replace the furnace and air conditioner with a heat pump.
- Replace the water heater with a heat pump water heater.
- Replace the gas stove with induction (when it dies, not before).
- Install solar.
- Buy an electric car (if you were buying a car anyway).
- Add a battery (last, and only sometimes).

Each step on this list does more for less money than the steps below it, and most of them do more than the steps below them by a lot.

The list is also almost the exact reverse of how most American households actually do it.

The reason for the reversal is not stupidity. It is a combination of three things.

First, **visibility**. Solar panels are visible. A new heat pump is invisible. So is insulation. If your goal is partly to signal your values to your neighbors — which is fine, by the way — solar wins automatically.

Second, **marketing**. There are tens of thousands of solar salespeople in America. There are maybe a few hundred dedicated air-sealing contractors. The industry that wants to sell you something is heavily tilted toward the things that are higher-margin and more visible.

Third, **incentives**. Federal and state programs have, historically, been heavier on solar tax credits than on insulation rebates. The Inflation Reduction Act of 2022 changed this somewhat, but solar still dominates the marketing.

The result is that a typical American clean-energy journey starts with a solar quote, includes some panels going up, and then the homeowner declares victory and goes back to having the same drafty walls and the same gas furnace they had before.

Let's walk through the right order, step by step.

The single best clean-energy investment most Americans can make in their homes has nothing to do with electricity. It has to do with the fact that their houses leak air like a screen door.

A typical older American house — say, anything built before 2000 — has gaps around windows, doors, plumbing penetrations, attic hatches, electrical outlets, and the rim joist where the foundation meets the framing. Through these gaps, in winter, warm indoor air leaks out and cold outdoor air leaks in. In summer, the opposite. The heating and cooling system runs harder to compensate. Every year, a significant fraction of the energy you pay for is being used to heat or cool the outdoors.

The fix is not exciting. A contractor with a blower door — a fan that pressurizes the house — finds the leaks. They get caulked or foamed shut. Attic insulation gets added. Maybe basement walls get sealed and insulated. The work takes a few days. It costs, depending on the house, anywhere from \$2,000 to \$10,000.

The payback is real and large. A weatherized house can use 20 to 30 percent less energy for heating and cooling than the same house unweatherized. Over the lifetime of the house, this is more than enough to pay back the cost — usually within five to ten years on energy bills alone.

And here is the deeper point. Every step *after* this one becomes cheaper and more effective when the house is properly sealed and insulated.

A heat pump for a leaky house has to be bigger and more expensive than a heat pump for a tight house. Solar panels for a leaky

house have to be more numerous, because the house is using more electricity. A battery for a leaky house has to be larger. Everything downstream is sized for the demand of the house, and the demand is set largely by how tight the house is.

This is why "seal and insulate first" is not just the cheapest step. It is the step that makes every other step less expensive.

It is also the step that is almost never on a homeowner's mind, because it does not show up on the roof.

If you have heard about one piece of clean-energy technology in the last few years that you should care about, this is the one.

A **heat pump** is a single device that replaces both a furnace and an air conditioner. In summer, it works like an air conditioner — moving heat from inside the house to outside. In winter, it works the same way *in reverse* — moving heat from outside the house to inside. Yes, even when it is cold outside, there is heat in the air. Cold air has less heat than warm air, but it has more than zero. The heat pump extracts that heat and moves it indoors.

The remarkable thing about a heat pump is that it does not *make* heat. It *moves* heat. A traditional gas furnace burns gas to create heat, getting maybe 90 percent of the energy in the gas converted to useful heating. A heat pump moves heat that already exists, and can deliver three or four units of heat for every unit of electricity it uses. This is called the **coefficient of performance**, or COP. A heat pump with a COP of 3.5 is, in effect, 350 percent efficient.

For your wallet and the climate, this means a heat pump in most American climates costs less to operate than a gas furnace, and produces significantly less carbon over its lifetime — especially as the grid gets cleaner.

The installation cost depends on the size of your house and the existing ductwork. A typical heat pump installation in 2026 costs \$10,000 to \$20,000 before incentives, and the IRA's tax credits and

rebates can knock thousands of dollars off that price for households that qualify.

A few things to know.

Modern heat pumps work well even in cold climates. Older heat pumps lost a lot of efficiency below freezing; newer "cold-climate" heat pumps can heat a house effectively down to -15°F or lower. Maine, Minnesota, and parts of Canada are now installing them faster than warmer states. The technology is no longer a southern-states-only solution.

A heat pump can be a single unit serving the whole house, or it can be installed as "mini-splits" — wall-mounted units in individual rooms that don't require ductwork. Mini-splits are useful for older houses where adding ductwork would be expensive.

A heat pump pulls more electricity than your old furnace blower did, because all the work that used to be done by burning gas is now being done electrically. This is why **a heat pump installation may require a panel upgrade**, especially in older houses with 100-amp service. We talked about that in Chapter 5.

The second-biggest energy load in a typical American home, after heating and cooling, is hot water.

A traditional water heater either burns gas to heat water in a tank, or uses electric resistance heating — which is just running electricity through a coil to make heat, the most wasteful way to use electricity for heating that has ever been invented.

A **heat pump water heater** uses the same principle as a regular heat pump. It pulls heat from the air around it (typically in your basement or utility room) and moves that heat into the water tank. The result is a water heater that uses about a third of the electricity of an electric resistance water heater, and that costs less to operate than most gas water heaters.

The cost is about \$1,500 to \$3,500 for the unit, plus a few hundred dollars to install. The IRA included generous rebates for these for low- and moderate-income households. Payback is typically five to ten years, sometimes faster.

A small caveat: a heat pump water heater takes longer to recover hot water after a big draw than a gas water heater does. If your family takes back-to-back showers in the morning, you may want a larger tank than you have now, or a slightly bigger unit. This is a manageable problem but worth knowing.

This one comes later in the list because it does not save nearly as much energy as the previous three. But it has two real benefits worth knowing.

Induction stoves are faster and more responsive than gas. They boil water in about half the time of a gas burner. They cool down the instant you turn them off. Many professional chefs now prefer them.

Gas stoves leak air pollutants into your home. Studies have found that gas stoves emit nitrogen dioxide, formaldehyde, and benzene at levels that, in some homes, exceed outdoor air quality standards. The link between gas stoves and childhood asthma is now well-established in the medical literature. The American Medical Association has called for action. Major appliance manufacturers are voluntarily reducing leak rates from new models, but the existing fleet of gas stoves in American homes — about 35 percent of all homes — is contributing to indoor air pollution every time it is used.

The case for replacing a gas stove with induction is therefore as much about your kids' lungs as about your carbon footprint. The case is real on both counts.

But: replacing a working stove is rarely cost-effective just to electrify. The right time is when the existing stove dies, or when you are renovating the kitchen anyway. Induction units cost \$1,000 to

\$3,000, plus possibly a 240-volt electrical run if the kitchen does not already have one.

Now, finally, we get to the visible thing.

By the time you have done steps 1 through 4, your house is using less electricity than it was before — possibly significantly less. The size of the solar system you need is smaller. The cost is lower. The payback is faster.

Everything in Chapter 9 still applies: orientation, roof age, shade, state policy (NEM 3.0 or its equivalent in your state), the local transformer, the panel inside. All of those constraints are still real.

But if your house has cleared all of those hurdles — if the roof is good, the policy is reasonable, the transformer can accept the solar — then a properly sized solar system after the other electrification is the right move. It can reduce your electric bill to nearly zero, or even produce a small credit, depending on rules.

The reason solar comes here, not first, is that **doing it first means oversizing the system for a house that is wasting energy**. You end up paying for a 10-kilowatt system to power a house that, if properly weatherized and electrified efficiently, would have only needed a 6-kilowatt system. The extra cost is real, and it is money you could have spent on the steps that actually reduce demand.

If you were going to buy a car anyway, get an electric one.

The carbon math on EVs has shifted decisively in the last five years. Almost everywhere in the United States, an EV produces less lifetime carbon than a gasoline car, even accounting for the emissions from manufacturing the battery. As the grid gets cleaner (Chapter 10, Chapter 11), the gap grows.

The reason "if you were going to buy a car anyway" matters is that the carbon and money cost of *replacing a perfectly good gasoline car* with an EV is much higher than the savings. The right time to

switch is when your current car needs to be replaced for normal reasons — age, mileage, accident, growing family.

A modern EV in 2026 costs \$30,000 to \$60,000, with federal and state credits available for many models. The total cost of ownership over ten years, including fuel and maintenance, is usually lower than a gasoline car of similar size.

An EV charging at home will add roughly 3,000 to 4,000 kilowatt-hours to your annual electricity use — enough to be noticeable on your bill but typically less than a heat pump for a similar-sized house. The same panel-upgrade concerns from Chapter 5 may apply.

Home batteries come last. We talked about this in Chapter 7, but it bears repeating in the context of an order-of-operations list.

A home battery makes sense if:

- You live somewhere with frequent outages.
- Your state has a time-of-use rate structure that pays well for storing solar.
- You have already done the other steps and have money left to spend.

A home battery does *not* make sense if:

- Your goal is purely to reduce your bill in a state with simple rate structures.
- You have not yet electrified your heating, water, or stove.
- You do not have rooftop solar yet (the battery is less useful without it).

The battery is the optional final step. Some households should never buy one. Others should buy one only after they have done everything else.

So why does the order matter so much?

It matters because the cheapest carbon reduction is, almost always, the one nobody can see. Insulation does not look like anything

from the street. A heat pump is a beige box behind the house. A heat pump water heater is a slightly taller cylinder in the basement.

The biggest impact you can make on your home's energy use will be invisible to everyone except your bank account and your electric meter.

This is the deeper lesson of this chapter, and one worth carrying past it. Many of the most important decisions in clean energy are about invisible things. The transmission line that does not get built. The transformer that gets upgraded. The insulation that gets added to a wall. The signal that nudges a thousand thermostats. These are the unglamorous moves that, in aggregate, decarbonize the world.

The solar panels on the roof get the attention. The wires in the wall do the work.

A note on incentives, which keep changing.

The Inflation Reduction Act of 2022 created the most substantial federal clean-energy incentive program in American history. It included tax credits and direct rebates for heat pumps, heat pump water heaters, insulation, weatherization, induction stoves, electric vehicles, rooftop solar, and home batteries.

The Trump administration, returning to office in 2025, has been working to roll back parts of the IRA. The full status of these programs as of 2026 is uncertain. Some tax credits remain. Some rebates have been suspended or restructured. State-level incentives continue to vary widely.

The practical implication: if you or your family is thinking about home electrification, **check the current programs in your state and at the federal level before making big decisions.** What was true when this book was written may not be true when you read it. The order of operations does not change with incentives. The *math* of which step pays back fastest can.

If federal incentives are reduced, the order of operations matters even more. Without subsidies, the cheapest-step-first logic becomes the only honest way to spend a household's clean-energy budget.

This chapter has been more practical than most. Less about the big invisible machines from earlier chapters, more about the choices a family might make over the next ten years.

The reason this practical chapter belongs in a book about the grid is that the grid is not just a thing that happens *to* you. It is increasingly a thing you are part of.

Every American home is, on the day this book is published, either an actively-participating node on the smart grid (smart meter, heat pump, solar, battery, EV) or a passive consumer of last-century electricity (gas furnace, gas water heater, old wiring, no solar). Most homes are somewhere in the middle.

The choice of where on this spectrum to be is partly economic, partly philosophical, partly about what your priorities are. There is no single right answer.

But if you are going to participate — if you have decided that electrification matters, that climate matters, that bill savings matter — then doing the steps in the right order will get you further per dollar than doing them in the wrong order.

If you are not yet a homeowner, this chapter is for the future you — the one who will be making these decisions in ten or fifteen years. The world will look different by then. The order of operations probably will not.

The most important things are often invisible. Start there.



CHAPTER FOURTEEN

The Trilemma



There is a question that has divided energy experts for fifty years, and it has an answer that most people who are not energy experts have never heard.

The question is this: what should we want from an electricity system?

The answer, on the surface, seems obvious. We want the lights to stay on. We want our bills to be affordable. We want the system to be clean — to not be wrecking the climate or poisoning the air.

The trouble is that no electricity system in the world delivers all three of these things equally well. Every country, every state, every utility, is making a trade-off. Sometimes the trade-offs are explicit, made by voters and regulators in public hearings. Sometimes they are implicit, baked into decisions made decades ago whose costs only become visible during a crisis.

This chapter is about the trade-off. It is about the framework that lets you understand why France's grid is different from Germany's grid is different from Texas's grid is different from California's grid. It is also about why understanding this framework will make you smarter than most adults about energy for the rest of your life.

The framework has a name. People who work in energy call it the **trilemma**.

Picture a triangle. Three corners. Three goals.

The first corner is **reliable**. The lights stay on. When you flip the switch, electricity is there. When a storm hits, the system recovers fast. There are no rolling blackouts during a heat wave. There are no surprises.

The second corner is **affordable**. The electricity bills are reasonable. Industry can compete on a global stage without being crippled by energy costs. Poor families can afford to heat their homes in winter. Schools and hospitals can run without breaking their budgets.

The third corner is **clean**. The system does not pump enormous amounts of carbon into the atmosphere. The air around power plants is breathable. Mining and waste impacts are managed responsibly. The lifecycle environmental damage is minimized.

Now here is the trilemma. **You can have any two of these three. You cannot have all three.**

If you optimize for reliable and affordable, you are mostly burning fossil fuels — particularly coal and natural gas. They are cheap, they are dispatchable, they are the easiest way to run a grid that never fails. But they emit carbon and pollute the air. So you are sacrificing clean.

If you optimize for clean and affordable, you are mostly building solar and wind. They are cheap, they are clean. But they are intermittent (Chapter 6), and storage is still expensive (Chapter 7). So you are sacrificing reliability unless you also build huge amounts of backup that you sometimes throw away.

If you optimize for reliable and clean, you are mostly building nuclear (Chapter 10), or nuclear plus storage plus oversized renewables plus a lot of transmission. You can do it. The lights stay on, the

carbon emissions stay low. But the system is *expensive*. You are sacrificing affordable.

This is not a thought experiment. It is what every country in the world is actually navigating right now.

To understand how the trilemma works in practice, it helps to look at four real systems.

In 1973, the world experienced an oil shock. The Yom Kippur War led Arab oil-producing countries to cut off shipments to the United States and parts of Europe. Within a few months, the price of oil quadrupled. Gas stations rationed fuel. Industrial economies that had spent the previous decade getting deeply addicted to cheap Middle Eastern oil found themselves staring at the bill.

The French government, which had less domestic oil and gas than most of its neighbors, made a decision that would shape the country for the next fifty years. France would, as a national policy, get most of its electricity from nuclear power. Quickly. Before the next oil crisis.

The decision was called the **Messmer Plan**, after Prime Minister Pierre Messmer. Over the next fifteen years, France built more than fifty nuclear reactors. By 1990, over 70 percent of French electricity came from these reactors — a fraction unmatched by any other country in the world.

The result, for France, has been remarkable.

French electricity has been among the cleanest in the developed world for decades. Per kilowatt-hour, French electricity emits roughly a tenth of the carbon of American electricity, and a fifth of German electricity. French air around power plants is some of the cleanest in Europe.

French electricity has also been remarkably reliable. The country has not had a major blackout in decades. The grid weathered the

2022 European energy crisis better than its neighbors — when Russian gas was cut off, France's nuclear reactors kept running.

But France paid a price. The construction of the nuclear fleet was enormously expensive. French electricity rates have historically been higher than in countries running on cheap coal or gas. The aging of the nuclear fleet is now a serious problem — most of those reactors are forty-plus years old, and they need expensive refurbishment or replacement. France's nuclear waste storage problem is, like everyone else's, unsolved.

When you hear that France "won" the energy game, the honest version is that France picked two corners of the trilemma — reliable and clean — and is still paying the bill for the third. The bill is real, and it is large. But France is also, fifty years later, one of the few major economies whose electricity is largely clean. Most of its neighbors cannot say the same.

The Messmer Plan was a bet. France won that bet, by most measures. But it was not a free win.

Germany made a different bet, decades later.

After the Fukushima nuclear accident in Japan in 2011, the German government — under Chancellor Angela Merkel — announced that Germany would phase out nuclear power. All of it. By 2022, every nuclear reactor in Germany would be shut down. This was on top of a separate German commitment to rapidly expand renewable energy, called the **Energiewende** — German for "energy transition."

The reasoning was a particular combination of post-Fukushima public fear, deep environmental movement skepticism of nuclear, and confidence that Germany could replace nuclear with solar, wind, and (importantly) imported Russian natural gas.

For a while, this seemed to work. Germany built more solar panels than any other European country. Wind farms went up across the north. Coal plants began to close. Total renewable share of

German electricity climbed from about 10 percent in 2005 to over 40 percent by 2020.

There were warning signs. German electricity rates climbed to among the highest in Europe — paying for the renewable buildout was expensive, even before you accounted for the cost of grid upgrades and storage. Germany's emissions did not drop as fast as expected, because closing nuclear plants meant burning more coal in the short term to make up the difference. Independent analysts pointed out that Germany was becoming dangerously dependent on Russian gas as a backup fuel.

Then, on February 24, 2022, Russia invaded Ukraine.

Within months, the Russian government cut off gas exports to most of Europe. German industry, which had been running on cheap Russian gas, suddenly faced energy costs three or four times higher than the previous year. Households faced staggering heating bills. The German government announced emergency measures — including, embarrassingly, restarting some of the same coal plants the Energiewende had been designed to close.

The Energiewende had, in essence, optimized for clean and (apparently) affordable while leaving reliability dependent on a foreign country whose intentions had been misread. When Russia changed the rules, all three corners of the trilemma got worse at once.

Germany is, several years later, slowly recovering. Renewable capacity continues to expand. Some nuclear plants have been kept running past their original shutdown dates. New gas terminals have been built to import liquefied gas from the United States and the Middle East. But the lesson is hard, and German voters have not forgotten it.

The Energiewende was an attempt to have all three corners of the trilemma. Many energy experts had warned, before 2022, that it was

relying on Russian gas to make the math work. They turned out to be right.

We covered Texas in Chapter 8 in the context of the February 2021 freeze. The honest framing through the trilemma is worth revisiting now.

Texas, for the last several decades, has optimized hard for **affordable**. Light regulation. An isolated grid that does not have to comply with federal reliability rules. Market structures that reward generators for the electricity they actually deliver, not for being available in emergencies. Cheap land for power plants. Easy permitting for new generation.

The results have been real. Texas electricity has, for most of the last twenty years, been cheaper than electricity in most other states. Texas has become the largest wind-power producer in the country, partly because the same easy regulatory environment that left it vulnerable to extreme weather also made it possible to permit and connect wind farms in months rather than years. Industrial customers love Texas electricity. Tech companies have flocked there partly for the cheap power.

Texas has also been, for most of those twenty years, fairly reliable. The 2003 Northeast blackout did not affect Texas. The 2021 freeze was an outlier — or at least, it was an outlier until February 2021. After February 2021, it was clear that the reliability had been borrowed against an extreme-weather bill that came due.

So Texas, in trilemma terms, picked affordable and (mostly) reliable. The clean corner is partial — Texas is the largest wind producer but also a huge natural gas user — and the reliable corner is conditional on weather staying within historical ranges.

After 2021, Texas has done some retrofitting. Winterization is now required by law. The grid has more battery storage than any other state. But the basic structural choices — isolated grid, light regula-

tion, market-based reliability — have not changed. Texas is still optimizing for affordable. The price is real.

California has spent the last twenty years running an aggressive experiment in clean energy.

The state passed laws requiring its utilities to generate 60 percent of their electricity from renewables by 2030 and 100 percent from clean sources by 2045. It paid generous subsidies to homeowners to install rooftop solar (until NEM 3.0 cut those subsidies — see Chapter 9). It built more battery storage than any other state. It pushed automakers, with state-level rules, to sell more electric cars.

The results, by some measures, are impressive. California now generates roughly half of its electricity from renewables during the daytime hours on most days. The state has more battery storage on its grid than any country except China. Solar panels cover roofs across the state in numbers that would have seemed impossible twenty years ago.

The results, by other measures, are sobering.

California has the highest residential electricity rates in the contiguous United States — about twice the national average. The reasons are complicated: wildfire-related infrastructure costs, transmission upgrades, paying for the early-build subsidies of the previous generation of renewables. Whatever the cause, ordinary Californians pay more for electricity than ordinary Americans almost anywhere else.

California has also had reliability scares. The 2020 rolling blackouts during a heat wave. The wildfire-related public safety power shutoffs. The duck curve from Chapter 6 — the daily challenge of evening ramps as solar fades. None of these are catastrophic in the way the Texas freeze was. But they are the reliability cost of pushing clean and affordable harder than most other states.

California, in trilemma terms, has picked clean and (mostly) reliable. The affordable corner has been sacrificed. Whether this is a worthwhile trade depends on what you value. California voters have, repeatedly, said yes.

There is a fifth system worth visiting in any honest discussion of the trilemma. It is the system the next chapter is entirely about.

China has built more solar than any other country, more wind than any other country, more transmission lines than any other country, more nuclear plants under construction than any other country, more electric cars than any other country, and — at the same time — more new coal plants than any other country. China is doing all of these things at once, at a scale and speed nothing else in human history compares to.

If you only looked at the showcase coastal cities — Shanghai, Beijing, Shenzhen — you might conclude that China has achieved all three corners of the trilemma. The lights are steady. Industry is humming. Air quality in those cities has improved dramatically in the last decade. Electricity rates are reasonable.

But, as the next chapter will explain, this picture is more complicated than it looks. Reliability in China's showcase cities exists in part because someone else, somewhere else in the same country, takes the cuts when supply runs short. The trilemma has not been solved. It has been *displaced* — onto people without political power to fight back.

We will get there. For now, the point is that the trilemma applies even to systems that look, from the outside, like they have somehow escaped it.

Here is what I want you to take from this chapter, more than any specific set of facts.

The trilemma is a tool. Once you have it, you can use it to evaluate almost any energy claim you hear for the rest of your life.

When someone tells you "France has the best energy policy in the world," ask: which two corners did France prioritize, and what did they pay for the third?

When someone tells you "Germany is a model for the energy transition," ask: clean and affordable — but reliable? Reliable how, exactly, when the wind stops blowing and the sun stops shining and the gas stops flowing?

When someone tells you "Texas has the cheapest electricity," ask: cheap on average, or cheap when you weight the cost by how it falls in February 2021?

When someone tells you "California is leading the world on clean energy," ask: leading at what cost, paid by whom, and how reliably?

The trilemma is also useful to apply to yourself. To your own utility. To your own state.

If your electricity rates are low, ask which corners your state has sacrificed.

If your electricity is clean, ask what you are paying for that.

If your grid is reliable, ask how much it costs and how clean it is.

There is no shame in any particular set of trade-offs. There is no universally right answer. But there are always trade-offs, and the people who pretend there are not are usually the ones trying to sell you something.

The trilemma applies to the United States as a whole, too. Not just to individual states.

For most of the twentieth century, the United States picked affordable and reliable. Electricity was cheap because we burned coal. Electricity was reliable because we built more capacity than we needed, with diverse sources, on a continental grid. The clean corner was sacrificed almost entirely.

In the twenty-first century, we have started trying to claw back some of the clean corner. We have built enormous amounts of natu-

ral gas (cleaner than coal but still fossil), solar, wind, and now nuclear. The clean corner is, slowly, being recovered. But we have not yet decided which of the other two corners we are willing to sacrifice to do it.

That decision — whether the United States will pay more for electricity, or accept a less reliable grid, in exchange for a cleaner one — is being made right now, mostly invisibly. Different states are making different bets. Different decades will prioritize different corners. The conversation about what we want from our grid is ongoing.

Most of the people making the choices are not consulting voters explicitly. They are state regulators, utility executives, environmental review boards, and federal agencies. The choices show up in your electric bill, in the air quality where you live, in the reliability of the grid during the next heat wave.

But the trilemma is the frame underneath all of it. If you understand the trilemma, you understand a lot of what is going on in the world.

You also understand that anyone who tells you it can be escaped is wrong.



CHAPTER FIFTEEN

The Lights in Shanghai



For nine years in Shanghai, I never experienced a blackout. Not one.

I lived in two apartments during those years. The first was visiting-faculty housing the university had assigned me when I arrived — an old walk-up, so old that the wiring inside the walls caught fire once. When the handyman came, I watched him drag a length of charred copper out of the plaster. The building was probably out of code. But even with burned wires inside the walls, the grid outside the walls never failed. The lights never blinked. The fridge never stuttered. The hallway bulbs came on at the same minute every evening as I climbed the stairs.

When I later moved to a newer rental across town — better building, modern wiring — the grid was still steady. Of course it was. It had been steady the whole time.

So I had a question I could not shake. Why, in a city of more than thirty million people, did the grid never once fail in nine years — even when the building it was feeding was a fire hazard? What was China doing that other countries were not?

The honest answer is one of the most important things this book can teach you, and it took me a long time to understand it.

There are real engineering achievements worth knowing about, and we should give them honest credit before we go any further.

Over the last twenty years, China built the world's longest and highest-voltage power transmission lines. These are called *ultra-high-voltage direct current* lines — UHVDC, if you want to sound impressive at dinner. They run more than three thousand kilometers across the country, carrying electricity from hydroelectric dams in the western mountains and wind farms in the northern grasslands all the way to the dense, hungry cities on the eastern coast. The wires are thicker than your arm. The towers are the size of small office buildings. No country has built anything like them at this scale.

China also built solar panels and lithium batteries faster than any country in history. In 2023 alone, China installed more solar capacity than the entire United States had built in all the years before. China makes about eighty percent of the world's solar panels. China makes most of the world's lithium batteries. They have the supply chain. They have the factories. They have the ports.

And they planned for it. China's central government decides what the country will build over five-year and ten-year horizons, and then it builds those things. When the government decides to put a transmission line across two thousand miles of territory, the line gets built. Land gets used. Villages sometimes get moved. There is not really a "no" anyone can say.

Those are real achievements. If you want to understand why the lights in Shanghai stayed on, you have to start by taking the engineering seriously.

But that is not the whole story. And the rest of the story is what this chapter is really about.

In the summer and fall of 2021, something happened in China that most Americans never heard about.

In a province called Guangdong, on the southern coast — home to about 130 million people, more than the entire population of Japan — the lights started going out.

Not all at once. Not everywhere. The pattern was specific.

Guangdong is where a huge amount of the world's manufacturing happens. Cities like Shenzhen, Dongguan, Foshan, and Guangzhou make phones, clothes, toys, electronics, furniture — most of the things you probably own. In May of 2021, twenty-one of those cities began rationing electricity to factories. By September, it was worse. Some factories were ordered to operate only two days a week. Some were told they could run one day a week. Some were cut off five days out of seven.

Sidewalks filled with diesel generators. Shopkeepers ran extension cords from the curb. Factory bosses, sitting in their offices by candlelight, told reporters they had not seen anything like it in a decade.

By the end of 2021, Guangdong had spent **sixty-six days** under power rationing. The province's own government later confirmed the number. It was the worst year for the Guangdong grid since at least 2010.

And it was not just Guangdong. Sixteen of China's thirty-one provincial regions ended up rationing electricity that year. In some northern provinces, the rationing reached homes — hospitals running on generators, traffic lights going dark, families losing heat in the early winter. In one frightening incident in the northeast, people were trapped in elevators when the power cut without warning.

This was happening at the same time my friends in Shanghai were going about their lives with no idea anything was wrong.

What caused it? The honest answer is that several things stacked on top of each other, the way things usually do when a system fails.

In late 2020, China had stopped buying coal from Australia after a political dispute. That removed a big source of the coal that fed Chinese power plants. Coal prices spiked.

The Chinese government had set the retail price of electricity by rule — utilities could only charge customers about 0.40 yuan per kilowatt-hour. But the cost of generating that kilowatt-hour, with expensive coal, had risen to about 0.60 yuan. Power plants were losing money on every unit of electricity they sold. Some of them simply stopped running.

Meanwhile, the southwestern province of Yunnan had a drought, which meant less water in the dams, which meant less hydroelectric power flowing east. The summer was hot. Demand was high. Air conditioners hummed across the country.

And on top of all of that, Beijing had set strict targets for emissions and energy use — a policy called *dual control*. Local officials, realizing they would miss their annual targets, abruptly ordered factories to shut down so the numbers would look better.

Stack those things together — the coal shortage, the price cap, the drought, the heat, the policy panic — and you get sixty-six days of darkness in the manufacturing heart of the country.

Here is the part that took me the longest to understand, and the part I most want you to remember.

When the Chinese grid runs short, it does not cut power randomly. It does not cut power proportionally — everyone losing a little. It cuts power according to a deliberate, designed order.

China Southern Power Grid, which runs the electricity in Guangdong and the surrounding provinces, said it openly in 2021. When supply ran short, they would prioritize *residential users, public utilities, and key clients*. Then, after that, ordinary industrial factories would be rationed.

That is the priority stack. Read it slowly:

Showcase megacities at the top — Beijing, Shanghai, Shenzhen, Guangzhou — protected hardest, because they are where foreign companies invest and where the country shows itself to the world. Then other residential users. Then "key clients," which usually means state-owned enterprises and politically important businesses. Then ordinary factories, last.

When the lights in my Shanghai apartment stayed on for nine years, it was not because the Chinese grid is magically more reliable than the American one. It was because Shanghai sits near the top of the priority stack. When supply got tight, somewhere else got cut first.

The factory worker in Dongguan, sent home for the week because her assembly line was ordered offline, was not having a separate experience from mine. We were having the *same* experience, viewed from opposite ends of the priority list. Her dark factory was the price of my unblinking lights.

This is what authoritarian capitalism — sometimes called state capitalism, sometimes national capitalism — actually looks like in practice. It is not just that the government can build big things fast. It is that the government can decide, in advance, who matters less when there is not enough to go around. The reliability you see in the showcase cities is real. The mechanism that produces it is also real.

So when someone — a politician, a tech CEO, a friend at dinner — tells you that "China is beating us on infrastructure" or "China is winning the energy race," there are two things you can hold in your head at once, without choosing between them.

The first is that some of what they say is true. China genuinely built the transmission lines the United States has been struggling to build for decades. China genuinely makes most of the world's solar panels. China genuinely plans further ahead than American utilities

are allowed to. There are things to learn here. The engineering is real.

The second is that the *reliability outcome* in Shanghai is not separable from the priority stack underneath it. You cannot copy the steady lights without also copying the choice about who gets darkness instead. The United States, with all its slowness and its arguments and its town meetings about where transmission lines can go, does not have that priority stack. A blackout in Manhattan and a blackout in rural West Virginia get treated with roughly the same urgency, because the regulatory structure is local and political accountability is local. That makes the American system slower to build. It also makes it harder to use as a tool of favoritism.

Both systems pay a price. Neither is winning. They are running different experiments, and the experiments cost different people different things.

Here is what I want you to take from this chapter. It is not really about China. It is a tool you can use for the rest of your life.

Whenever someone tells you that another country has better infrastructure, or better healthcare, or better trains, or better internet, the right next question — the one most adults will not ask, and the one that will make you smarter than most adults in the room — is:

Better for whom?

That question travels. Use it on every comparison you hear. Better infrastructure for whom? Better hospitals for whom? Better schools for whom? Most comparisons between countries collapse the moment you ask. Some get stronger. Either way, you learn something real.

For nine years, the lights in my Shanghai apartment did not blink. It took me longer than it should have to understand why. I hope it takes you less time than it took me.



CHAPTER SIXTEEN

The People Who Watch the Grid



It is four in the morning at a regional electric utility somewhere in the eastern United States. Outside, the sky is dark. Inside the control room, the wall of screens from Chapter 2 still glows. The numbers still scroll. The frequency on the central display reads 60.000.

A woman in her fifties is sitting in the chair. Her name is not important — there are people like her in every control room in the country — but her job is one of the most important in modern life. She is a system dispatcher. For the next eight hours, she is responsible for the moment-to-moment balance of an electrical grid that serves several million people.

She has been doing this job for seventeen years. Before that, she worked at a coal plant in Kentucky, running the boiler. Before that, she was in the Navy as an electrician's mate aboard a nuclear submarine. The combination of nuclear training, plant experience, and grid dispatching is common in this profession. The skills transfer.

At 4:12 AM, a generator at a coal plant in West Virginia trips offline. The first indication is a small wobble in frequency — the heartbeat dipping from 60.000 to 59.997 — and a red alert on her screen. Within three seconds, the grid's automatic generation control

begins ramping up other plants to compensate. She watches the frequency recover. She sends a message to the next shift's morning briefing logbook. She drinks her coffee.

This is the unsung work of the modern world. It is happening right now, somewhere, as you read this.

This chapter is about the people who do it.

For most of the last fifteen chapters, we have been describing the electrical grid as a machine. A juggling act. A synchronized continent-sized machine. A system with a brain.

These metaphors are all useful. But they obscure something important.

The grid is not actually a machine, in the way a car or a microwave is a machine. The grid is a service that exists because tens of thousands of human beings, every day, make it exist. The hardware is essential — the generators, the wires, the transformers, the substations. The software is essential — the energy management systems, the alarm software, the demand-response platforms. But the hardware and software do not run themselves. They are operated, maintained, repaired, planned, forecasted, and defended by people.

These people are mostly invisible. You will never meet most of them. You probably do not know what their job titles are. Their work, when it is going well, is so smooth that you forget it is happening. Their work, when it goes badly, becomes a headline for a few days and then disappears.

This chapter is a brief tour. The book has met some of these people already — the office boy in 1979, the FirstEnergy operators in 2003, the dispatcher at PJM in Audubon, the Texas grid operators who came within four minutes of catastrophic collapse. It is worth, before we close, devoting a chapter to the people themselves rather than to the events they were characters in.

There are a few roles especially worth knowing about.

The dispatchers are the people at the wall of screens.

Most large American grids — the eastern interconnection, the western, ERCOT in Texas — are run by regional organizations called Independent System Operators or Regional Transmission Organizations. The biggest is **PJM**, headquartered in Audubon, Pennsylvania, serving sixty-five million people across thirteen states and the District of Columbia. Others include the Midwest Independent System Operator (MISO), the California Independent System Operator (CAISO), the New York Independent System Operator (NYISO), and ERCOT.

Each of these organizations runs control rooms that operate around the clock, every day of the year. Christmas morning. New Year's Eve. The middle of a hurricane. The afternoon of the Super Bowl. The grid never sleeps, and the dispatchers do not either.

The path into the job is rarely direct. Many dispatchers come from the military, particularly the Navy's nuclear program — the training is rigorous and similar in style. Many come from plant operations, having spent years running coal, gas, or nuclear plants before moving to the more abstract work of the control room. A few come from engineering programs at universities. The federal government requires every dispatcher to be certified by the North American Electric Reliability Corporation (NERC), with continuous education and recertification every few years.

The job is high-pressure. Most days are quiet. But on the days that are not — a heat wave pushing the grid to its limits, a storm taking out transmission lines, a generator tripping unexpectedly — the dispatchers have minutes, sometimes seconds, to make decisions that affect millions of people. They train for these days the way pilots train for emergencies: in simulators, repeatedly, with deliberate stress.

Most dispatchers I have ever read about or heard described love their work. They are deeply protective of the grid. They get angry when politicians talk about it carelessly. They roll their eyes at marketing language. They watch each other's careers across the industry. When a dispatcher retires after thirty years in the chair, there is usually a quiet ceremony.

They are the closest thing the modern world has to lighthouse keepers. Most people will never know their names.

If the dispatchers are the calm ones, the lineworkers are the rough ones.

A **lineworker** — sometimes called a lineman, though the workforce now includes women — is the person who climbs the utility poles. They are the ones in the bucket trucks on the side of the highway. They are the ones who go up in storms.

The work is physical. It is dangerous. It pays well, by trade standards — a journeyman lineworker in the United States typically makes \$80,000 to \$130,000 per year, depending on overtime and storm work. It requires years of apprenticeship. It involves working with voltages that can kill you instantly. Lineworkers have one of the highest fatality rates of any American trade — roughly 20 deaths per 100,000 workers per year, several times the national average.

When a hurricane hits the Gulf Coast or a snowstorm shuts down New England, the lineworkers from neighboring states drive in. Mutual aid agreements between utilities allow crews from Texas to help in Florida, crews from Michigan to help in New York, crews from Ohio to help in Vermont. The driving alone can take days. The work, once they arrive, can take weeks. Lineworkers sleep in shifts in hotels or in tents at staging grounds, eat catered meals, and work twelve-hour days in conditions that range from miserable to dangerous.

After Hurricane Maria devastated Puerto Rico in 2017, lineworkers from across the continental United States flew in to help restore the grid. The work continued for months. Some of those crews are still talked about today in the industry — both for the heroism of the work and for the moments when the systems of mutual aid broke down and crews were stranded or under-supported.

When you see a power-restoration update on the news after a storm, the number of customers restored each hour is a statistic. The actual work is being done by a person, on a pole, in the dark, while the storm is still receding. You will probably never meet them. They will probably restore your power one day anyway.

The dispatchers tell the plants what to do. The **plant operators** actually do it.

A coal plant, a gas plant, a nuclear plant — each is a complicated industrial facility, full of pipes and pumps and turbines and electrical equipment. Running one safely requires teams of people working continuously, monitoring temperatures and pressures and flow rates, making small adjustments, performing maintenance, responding to alarms.

Nuclear plant operators are a category of their own. The training to become a licensed reactor operator in the United States takes about two years, full-time, and includes extensive coursework, simulator training, and supervised on-the-job experience. The Nuclear Regulatory Commission examines and re-examines operators throughout their careers. A licensed senior reactor operator at a US nuclear plant is one of the most credentialed industrial professionals in the country.

Coal plant operators are less famous but no less skilled. Modern coal plants are enormously complex machines that, when running well, operate at high efficiency for years at a time. The recent decline of coal in the American grid means that many of these operators are

being retrained or retired — and the institutional knowledge of running large thermal plants is, slowly, being lost. This matters more than it sounds. As the grid restructures, we may need that knowledge again, or we may not. Either way, it is leaving faster than it is being replaced.

Gas plant operators, especially at the new generation of combined-cycle plants, are doing some of the most technically interesting work in the industry. A modern gas plant can ramp from idle to full power in twenty or thirty minutes. The operators are flying a complicated machine through rapid changes — exactly the flexibility the grid needs to handle the duck curve from Chapter 6.

A **transmission planner** is an engineer who thinks about the grid five to twenty years in the future.

This is one of the most thankless jobs in the entire industry, and one of the most important. The grid we have today is mostly the result of decisions made fifteen to forty years ago by people whose names you will never know. The grid we will have in 2050 is being designed right now, by another generation of people whose names you will also never know.

A transmission planner studies load forecasts (where will demand grow?), generation queues (where will new power plants want to connect?), and constraint maps (where is the existing grid running out of capacity?). They model the grid under thousands of hypothetical conditions — what if a particular line fails on a hot day? What if two plants trip at the same time? What if there is a sudden surge of EVs being charged at 6 PM? They identify where new lines and substations need to be built, and they propose those projects to regulators and to neighboring utilities.

Their work is mostly invisible until something goes wrong. A blackout caused by a missing transmission line is a planning failure. A bottleneck that delays interconnection of new renewables is a

planning failure. The 2,500 gigawatts of generation projects stuck in interconnection queues globally (Chapter 11) reflect, in part, the limits of how fast transmission planning can move.

Transmission planners are also, increasingly, in the middle of political fights. The Northern Pass project from Chapter 3, the SunZia project, the failed Plains and Eastern transmission line in Oklahoma — every major transmission battle has, somewhere in it, a team of planners whose careful work has been overtaken by political opposition or by litigation. Some of them retire frustrated. Most of them stay, and start another project.

If you are reading this and you like engineering and you also like long-term thinking, this is an underappreciated career path.

Weather drives electricity demand. Weather drives renewable generation. Therefore weather forecasters are now, more than ever, part of the grid.

A **forecaster** at a major utility or grid operator is doing two things at once: predicting consumer demand (how many air conditioners will be running in seven days?) and predicting renewable supply (how much wind will be blowing in fourteen hours? how much sun?). The accuracy of both forecasts directly determines how the grid is operated.

For decades, this work was done by traditional meteorologists with traditional weather models. The forecasts were good but not great. Misses cost the grid money, sometimes a lot of money.

In the last ten years, forecasting has been transformed by machine learning. Modern grid forecasts use neural networks trained on enormous data sets of past weather, past demand, and past renewable output. The result is forecasts that are dramatically more accurate than they were in 2015 — and that get better every year.

This is one of the quiet ways AI is improving the grid that almost nobody talks about. The same machine learning that lets a phone

recognize your voice or a website recommend a movie is also, today, telling grid operators how much wind to expect in Texas tomorrow afternoon. The forecasters who survived the transition learned to work with the algorithms. The ones who did not have mostly moved on.

The newest role on this list did not exist forty years ago, and is now one of the most important.

A **cybersecurity** professional at an electric utility is defending against attacks that, if successful, could darken cities and damage equipment. The 2015 attack on the Ukrainian grid was the first publicly known case of a state-sponsored cyberattack actually causing a blackout. Russian hackers reached into a Ukrainian utility's control system, opened circuit breakers, and left about 230,000 people without power for several hours. The 2016 follow-up attack was more sophisticated. Both events are studied at every utility in the developed world.

American utilities have, so far, avoided major successful attacks. This is not by accident. The NERC Critical Infrastructure Protection standards — the CIP rules — require utilities to harden their networks, segment their systems, monitor for intrusions, and respond to incidents. The teams that do this work are usually invisible. When they succeed, nothing happens. When they fail, history changes.

The cybersecurity workforce is the youngest part of the grid industry. It is also one of the hardest to staff. Talented cybersecurity professionals can make more money in finance or tech than they can at utilities, and the work is often more glamorous. Recruiting and retaining the people who keep the grid digitally safe is, quietly, one of the biggest workforce challenges in the industry.

One more role, the most mundane, the most unglamorous, and — after 2003 — the most legally required.

Tree-trimming crews are the people who walk the transmission and distribution corridors with chainsaws and chippers, keeping the right-of-way clear of vegetation. After the 2003 blackout (Chapter 4), federal regulators imposed mandatory tree-trimming schedules on every transmission owner in the United States. Failure to comply now carries fines of up to a million dollars per day per violation.

The work is hot, dirty, and dangerous. Falling branches kill workers. Power lines kill workers. The hours are long. The pay is decent but not high.

Most people never think about tree-trimming crews. You may pass them on the side of a road and not notice. You should notice, though. They are the last line of defense between an aging American grid and the natural world that constantly threatens to overgrow it.

There is one more thing to say about all of these people, and it is the most important.

The American grid workforce is aging.

The average lineworker is over forty-five. The average dispatcher is older. Many transmission planners are in their fifties and sixties. The retirement wave that has been building for years is now arriving, and the new generation is not being trained fast enough to replace it.

This is a real and underappreciated risk for the modern grid. All the new technology in the world — the smart grid, the SMRs, the virtual power plants, the AI forecasting — requires people to operate it. And those people require years of training, much of which is delivered by the experts who are now retiring.

There are bright spots. Some utility apprenticeship programs are growing. Community colleges in places like New England and the Pacific Northwest are training lineworkers in larger numbers than they were a decade ago. The Department of Energy has launched several workforce initiatives. But the gap between people retiring

and people entering the field is real, and it is unlikely to close on its own.

If you are reading this and thinking about what to do with your life, the American grid is in genuine need of new people, at every level. Engineers, electricians, dispatchers, planners, forecasters, cybersecurity professionals, lineworkers, mechanics. The jobs are real. The pay is decent. The work is meaningful. The training programs exist.

It is a not-glamorous secret of the next twenty years that the people who keep the lights on will, increasingly, be in short supply.

The office boy in Chapter 1 had a name that I do not know. The dispatchers in Audubon do too. The FirstEnergy operators in 2003. The lineworkers in Puerto Rico after Maria. The New York City police officers who climbed down into the subway tunnels in August 2003 and led the people in suits and heels back up to the sun.

There are tens of thousands of these people working today. They will not show up in your news feed unless something goes terribly wrong. They will not be at the front of the next big tech keynote. They will, on a quiet Tuesday afternoon in November, be in a control room you have never seen, watching a number — 60.000 — and listening for the moment something changes.

This entire book has been, in a way, about them. Every machine in every previous chapter is operated, in the end, by a person. Every blackout that did not happen was prevented by someone whose job was to prevent it. Every restoration after a storm was accomplished by someone on a pole.

If you are looking up at a power line today, somewhere a person climbed up to put it there. If your lights are on tonight, somewhere a person is making sure they stay on.

The grid is, ultimately, a service that exists because human beings keep it existing. The technology can change. The economics can

change. The fuels can change. The framework can change. The human beings, at the center of it all, do not.



CHAPTER SEVENTEEN

The Grid You'll Build



If you are reading this book in a house with electricity, the grid is, right now, working for you.

A turbine is spinning somewhere. A wire is humming somewhere. A dispatcher is watching a number on a screen somewhere. A transformer is converting voltage somewhere. A control room is quietly running. The lights are on. You probably are not thinking about any of it.

If you are reading this book in 2030, much of what you just learned is still true. The grid is still mostly the same machine, still mostly running on the same physics, still mostly operated by the same kinds of people. Some new things have been built. Some old things have been retired. The basics have not changed.

If you are reading this book in 2040, more has changed than you might expect. The reactors at the Crane Clean Energy Center are humming. Several small modular reactors have been operating for a decade. Several have failed. The duck curve has been mostly tamed by storage in some states and not yet tamed in others. Loudoun County's data centers are several times bigger than they are now. The XFRA pilots have either become routine or quietly disappeared.

If you are reading this book in 2050, the grid will be a thing your generation built.

This chapter is about that.

Let's take a quick inventory of what we have learned.

The grid is one giant synchronized machine. Electricity is made and used in the same instant. The frequency is the heartbeat — 60 Hz — and it tells you whether supply and demand are in balance. Dispatchers in control rooms watch the heartbeat, every minute of every day.

Transmission lines move electricity from where it is made to where it is used. They have to be enormous because the physics of resistance demands high voltage for long distance. Building new ones takes too long because of permitting, not technology. The same is true of substations, distribution upgrades, and almost every other piece of physical grid infrastructure.

Renewable energy is now the cheapest electricity per kilowatt-hour in most of the world. But it is intermittent, and a grid cannot run on intermittency alone. Storage helps. Nuclear helps. Transmission helps. None of these is fully solved at the scale required.

The grid is becoming smarter. Sensors, smart meters, demand response, distributed energy resources, virtual power plants — all of these are real, partially deployed, and growing. Some homes are now power plants. Some are now data centers.

Different countries have made different bets on the trilemma. None of them has all three corners. Each has paid different prices. The conversation about which corner to sacrifice is ongoing in every state in the United States and every country in the world.

And underneath all of it, tens of thousands of people are doing the actual work. Most of them are graying. Most of their successors have not yet been trained.

That is the picture as of 2026. It will change.

There are, in 2026, a handful of unsolved problems that the next twenty years of energy work will mostly be about.

These are not minor. They are the problems that will determine whether the world can build a grid that is genuinely clean, genuinely affordable, and genuinely reliable — or whether some corner of the trilemma will continue to be sacrificed because we did not figure out how to do better.

I want to name them, because the people who solve them will probably be reading this book. Or books like it. Or being taught by people who read this book.

Long-duration storage. Lithium batteries handle four hours. Pumped hydro handles a day, but only in places with mountains. We do not have an economical way to store electricity for a week, let alone a season. Without long-duration storage, a grid running mostly on solar and wind cannot reliably handle a windless cloudy week in February. Iron-air batteries, hydrogen, compressed air, thermal storage, gravity storage — all of these are candidates. None has yet been deployed at the scale required. Whoever solves this gets to redesign the grid.

Transmission permitting. The wires we need to build between now and 2050 are wires that have not yet been planned, on land that has not yet been crossed, through political and legal fights that have not yet been won. The technology is not the bottleneck. The bottleneck is the seventeen-year permitting timeline of the SunZia project, the eight-year fight over Northern Pass, the lawsuits and counter-lawsuits and environmental reviews of every major line proposed in the last twenty years. Somebody has to figure out how to build new transmission corridors in years rather than decades — without trampling the communities, ecologies, and property rights that have legitimately resisted them in the past.

Grid-forming inverters. The synchronized machine from Chapter 2 has, for a hundred years, been held together by the inertia of massive spinning generators. Solar and battery inverters do not have inertia. Engineers are working on inverters that can mimic inertia and even stabilize a grid without any spinning mass at all. The technology is improving fast but not yet deployed at scale. A grid that can run on solar, wind, and batteries alone — without any rotating thermal plants — is not yet engineering reality. It will be.

Demand-side coordination at scale. The smart grid is partially deployed. Demand response programs exist. Virtual power plants exist. But the system that would coordinate billions of small assets — every thermostat, every EV charger, every water heater, every Powerwall — in real time, across the entire country, with appropriate privacy and security safeguards, has not been built. Most of the technology exists. The systems engineering does not.

Cybersecurity. A digital grid is a hackable grid. The Ukrainian attacks in 2015 and 2016 showed what is possible. The defensive work is ongoing, and the attackers are also getting better. This will not be solved once. It will be a permanent arms race for the rest of the century.

Workforce. Every problem above requires people to solve it. The workforce is graying faster than it is being replaced. Closing the gap is going to require apprenticeships, community colleges, university engineering programs, and a cultural shift in how Americans think about working on physical infrastructure. The pay is decent. The work is meaningful. The training programs exist or can be built. The bottleneck is whether enough young people choose to do this work.

Cost-effective firm clean power. The deepest open problem. Nuclear is firm and clean but expensive. Geothermal is firm and clean but limited to certain geologies, at least so far. Long-duration storage might solve it but is not yet proven at scale. Carbon capture for gas

plants might solve it but is not yet economic. We need a source — or some combination of sources — that delivers reliable electricity 24 hours a day with low carbon emissions at competitive cost. We do not yet know which technology will win. Whoever figures it out will, in the most literal sense, build the grid of the 2050s.

If you are reading this book in your teens, you are looking at a generational opportunity.

The grid your generation will inherit is being designed, right now, by people who will mostly be retired by the time you are running it. The choices being made today — to build this gas plant or that nuclear reactor, to permit this transmission line or block it, to invest in batteries or hydrogen, to upgrade the smart grid or hold back — will be your inheritance.

But the choices being made *next* decade will be the ones you make.

You do not have to become an electrical engineer to be part of this. The grid needs lawyers who specialize in permitting reform. It needs policy experts who can navigate state utility commissions. It needs entrepreneurs who can build the companies that will deploy the technology. It needs journalists who can explain to the public what is actually happening. It needs financial people who can structure the deals that fund the infrastructure. It needs urban planners who can think about land use in the era of distributed energy. It needs writers who can tell the stories that change minds.

It also, very much, needs the engineers — the chemists working on iron-air batteries in Massachusetts, the nuclear engineers building Hermes reactors in Tennessee, the software architects writing the next generation of grid management systems, the meteorologists training AI to forecast wind speeds eighteen hours out. Those jobs are growing faster than they can be filled.

The grid your generation builds will be different from the one you grew up on. It will probably be cleaner, almost certainly more digital, possibly more distributed, definitely more contested. It will not be perfect. The trilemma will still apply.

But it will, in significant ways, be yours.

Let me close where this book started.

In Chapter 1, I told you about the Blink — the small flicker of my Hoboken apartment's bulbs every time the air conditioner turned on. I told you that the Blink contained, in miniature, the same physics as the 2003 Northeast blackout. The same balance act. The same juggling of supply and demand, in a fraction of a second.

You have, since then, met the synchronized continent-sized machine that produced the Blink. You have walked through transmission lines and substations and pole transformers. You have read about the duck curve and the lithium batteries that are starting to flatten it. You have heard the story of the FirstEnergy operators in 2003 and the ERCOT operators in 2021. You have walked along the Susquehanna River and watched the cooling towers prepare to come back to life. You have stood in Loudoun County and looked at the office park where so much of the modern world's data lives. You have learned the trilemma, and you have learned to apply it to any energy claim you ever encounter.

You have also met the people. The dispatchers. The lineworkers. The plant operators. The transmission planners. The forecasters. The cybersecurity teams. The tree-trimming crews. The office boy who drove everyone home on August 14, 2003.

So now, when you flip the switch and the lights come on, you know what is happening on the other side. A turbine somewhere is spinning. Wires somewhere are humming. A dispatcher somewhere is watching a number. A transformer somewhere is doing its small

piece of the work. Tens of thousands of people, most of whom you will never meet, are keeping the grid alive for one more minute.

You also know that this is not stable. The grid is being rebuilt. The technologies are changing. The fuels are changing. The financial structures are changing. The workforce is turning over. The trilemma is being re-negotiated.

Pay attention next time you flip a switch.

The whole world is on the other side of it.

Some of it works because of choices made a hundred years ago. The rest is being decided now, by people not much older than you are.

And the rest of the rest will be decided by you.

